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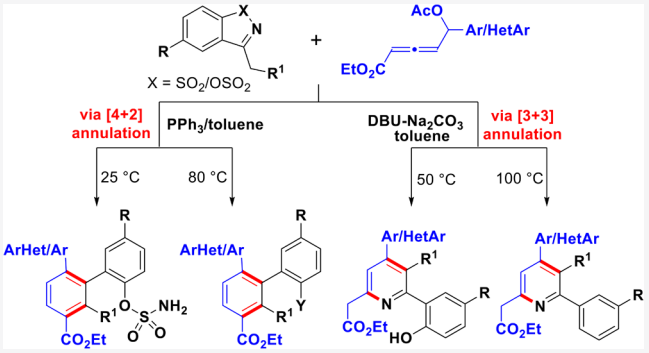
Lewis Base-Switched [3 + 3] and [4 + 2] Annulation Reactions of δ-Acetoxy Allenoates with Cyclic *N*-Sulfonyl Imines: Divergent Synthesis of Functionalized α-Pyridyl Acetates and Teraryl Scaffolds

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ABSTRACT: Under metal-free conditions, δ-acetoxy allenoates react with cyclic *N*-sulfonyl imines (sulfamidate imines/sulfonyl imines) to afford functionalized 2-pyridinyl acetates (α-pyridyl acetates) or teraryl motifs by a simple Lewis base switch. Thus, while DBU/Na_2CO_3 combination-directed [3 + 3] annulation involves sulfonyl elimination via O−S or C−S bond cleavage, affording 2-pyridinyl acetates, Ph_3P-catalyzed [4 + 2] annulation leads to functionalized teraryls via Mannich coupling and C−N bond cleavage with retention or cleavage of the sulfamoyloxy group depending on the reaction conditions.



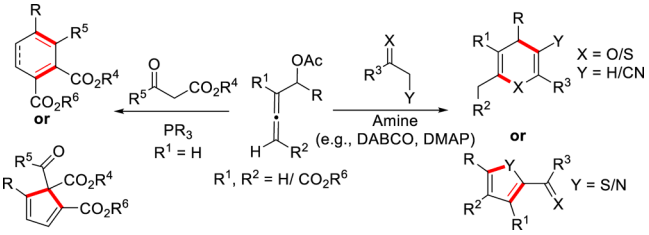
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INTRODUCTION

Allenes have been well recognized as key building blocks in organic synthesis.[1] During the past decade, the Lewis base-catalyzed annulation of allenoates has emerged as one of the most powerful tools for the synthesis of structurally divergent carbo- and heterocyclic motifs.[2] Subsequent to the early example of phosphine-catalyzed (3 + 2) annulation of butadienoate with electron-deficient olefins reported by Zhang and Lu,[3] research on the Lewis base-catalyzed annulations of a variety of allenoates have been extensively studied. For example, phosphine-catalyzed [3 + 2],[4] [4 + 1],[5] [4 + 2],[6] and [4 + 4][7] cycloaddition reactions were developed as effective approaches for the construction of a wide variety of carbo/heterocyclic rings and medicinally significant natural compounds.[8] In addition to phosphines, tertiary amines have also been extensively investigated as nucleophilic catalysts to facilitate numerous annulation reactions of allenoates. For instance, [4 + 2], [3 + 3], and other annulations of allenoates have been reported for the assembly of heterocycles.[9] Moreover, β′/δ-acetoxy allenoates have also been identified as effective annulation partners under Lewis base catalysis by Tong as well as others (cf. Scheme 1).[10] An important point to be noted in these organocatalytic reactions is the difference between phosphines and amines as organocatalysts leading to distinctly different products in several cases.[11]

The α-pyridyl acetates as well as teraryl scaffolds with hydroxyl/sulfonamide functionality have been found in a variety of natural and pharmaceutical compounds.[12] These scaffolds are also explored as versatile intermediates to construct complex molecules.[13] A literature survey reveals

Scheme 1. Reactions of δ-Acetoxy Allenoates for the Synthesis of Carbo- and Heterocycles



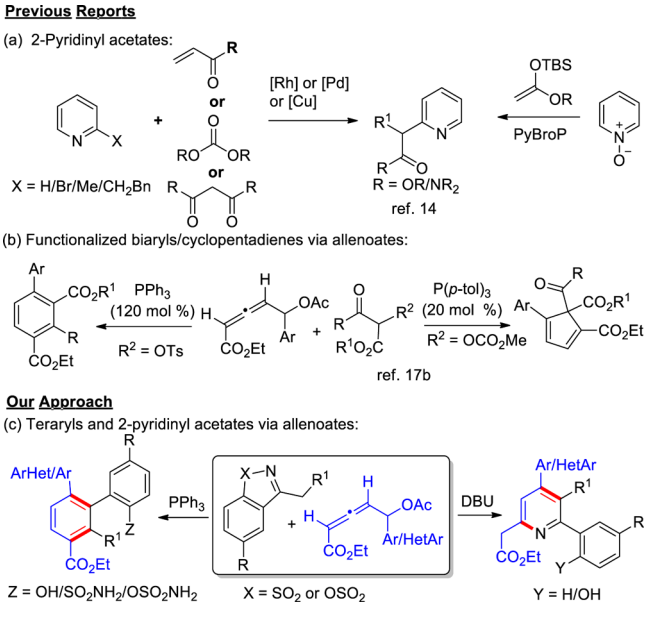
that a fair amount of research is focused on α-pyridyl acetates (cf. Scheme 2a)[14] and teraryls.[15] However, these strategies have several drawbacks such as the requirement of expensive transition metal catalysts, high temperature, unstable reagents, and strong bases. Hence, a less expensive metal-free approach for the synthesis of such structural motifs is in urgent demand. Despite the substantial progress made on acetoxy allenoates, there is no report of annulation reaction between δ-acetoxy allenoates and bi-nucleophilic cyclic *N*-sulfonyl imines.[16] A combination of acetoxy allenoates and cyclic *N*-sulfonyl imines may lead to 2-pyridinyl acetates and unsymmetrical *o*-teraryls

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Scheme 2. Formation of 2-Pyridinyl Carboxylate and Teraryl Scaffolds

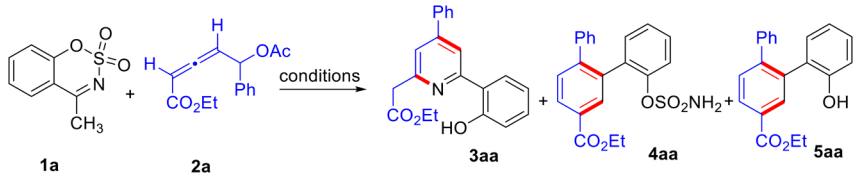


with a hydroxyl, sulfonamide, or sulfamoyloxy unit, which are otherwise difficult to obtain by reported procedures. A recent elegant report from Tong's group disclosed the synthesis of multisubstituted biaryl motifs from δ-acetoxy allenoates that requires a tosyloxy group at the second position of β-keto ester and use of Ph_3P (Scheme 2b).[17] In this context, our interest in allene chemistry[1f,18] motivated us to explore these substrates for new Lewis base-switched annulations. It may be noted that sulfamidate imines/sulfonyl imines can act as nucleophiles (active methylene group) as well as electrophiles (imine moiety). Our work reported herein embodies divergent annulations of δ-acetoxy allenoates with cyclic *N*-sulfonyl imines (sulfamidate imines/sulfonyl imines) under metal-free conditions and involves a simple Lewis base switch to afford (i) a new class of α-pyridyl acetates (2-pyridinyl acetates) by [3 + 3] annulation via SO_2 elimination and (ii) functionalized *o*-teraryls by [4 + 2]-benzannulation through C−N bond cleavage (Scheme 2c). While the reaction with sulfamidate imines involves O−S bond cleavage, the one with sulfonyl imines entails C−S bond cleavage.

RESULTS AND DISCUSSION

Initially, 4-methyl cyclic sulfamidate imine **1a** and δ-acetoxy allenoate **2a** were chosen as model substrates. As depicted in Table 1, we tested the reaction of **1a** (1.0 equiv) with **2a** (1.2 equiv) by using DABCO (0.20 equiv) in toluene at rt (25 °C). However, only a trace amount of desired α-pyridyl acetate **3aa** was observed (entry 1). The yield could be enhanced to 31% by using 50 mol % DABCO (entry 2). Addition of Na_2CO_3 (1.0 equiv) improved the yield of **3aa** to 43% (entry 3); further increase to 56% was achieved by performing the reaction at 50 °C (entry 4). Use of 1.5 or 2.0 equiv Na_2CO_3 augmented the yield to 62% (entries 5 and 6). Among the bases (DBU, Et_3N, DIPEA, and DMAP) checked, DBU turned out to be the best (entry 7, also see Table S1). The additives K_2CO_3 and *t*-BuOK in place of Na_2CO_3 produced **3aa** in slightly lower yield (entries 8 and 9). Among the screened solvents (cf. Table S1), toluene was the best, affording the desired product **3aa** in 79% yield (entry 7). There was no quantifiable difference in the isolated yields with increased time (entry 10). Interestingly, use of PPh_3 (TPP) in place of DBU provided the entirely different product **4aa** (53%; entry 11). To increase the yield of the latter product **4aa**, we performed the reaction between **1a** and **2a** at different temperatures. Surprisingly, the S−O bond-

Table 1. Optimization of Reaction Conditions for 3aa, 4aa, and 5aa[a]



entry	base	additive	solvent	*T* (°C)	yield (%)[b] (3aa:4aa:5aa)
1	DABCO		$PhCH_3$	25	<5:00:00
2	DABCO		$PhCH_3$	25	31:00:00
3	DABCO	Na_2CO_3	$PhCH_3$	25	43:00:00
4	DABCO	Na_2CO_3	$PhCH_3$	50	56:00:00
5	DABCO	Na_2CO_3	$PhCH_3$	50	62:00:00
6	DABCO	Na_2CO_3	$PhCH_3$	50	62:00:00
7	DBU	Na_2CO_3	$PhCH_3$	50	79:00:00
8	DBU	K_2CO_3	$PhCH_3$	50	72:00:00
9	DBU	*t*-BuOK	$PhCH_3$	50	75:00:00
10	DBU	Na_2CO_3	$PhCH_3$	50	79:00:00
11	PPh_3	Na_2CO_3	$PhCH_3$	50	00:53:00
12	PPh_3	Na_2CO_3	$PhCH_3$	80	00:00:73
13	PPh_3		$PhCH_3$	80	00:00:81
14	PPh_3		$PhCH_3$	25	00:73:00
15	PPh_3		$PhCH_3$	80	00:00:81

[a]Reaction conditions: **1a** (0.20 mmol) and **2a** (0.24 mmol) with base (20 mol % for entries 1 and 13−15; 50 mol % for entries 2−12) and additive (1.0 equiv for entries 1 and 2 and 1.5 equiv for entries 5−10) in toluene (2.0 mL); temperature is that of oil bath. [b]Isolated yield (after 12 h for entries 1−9 and 11−14; after 24 h for entries 10 and 15).

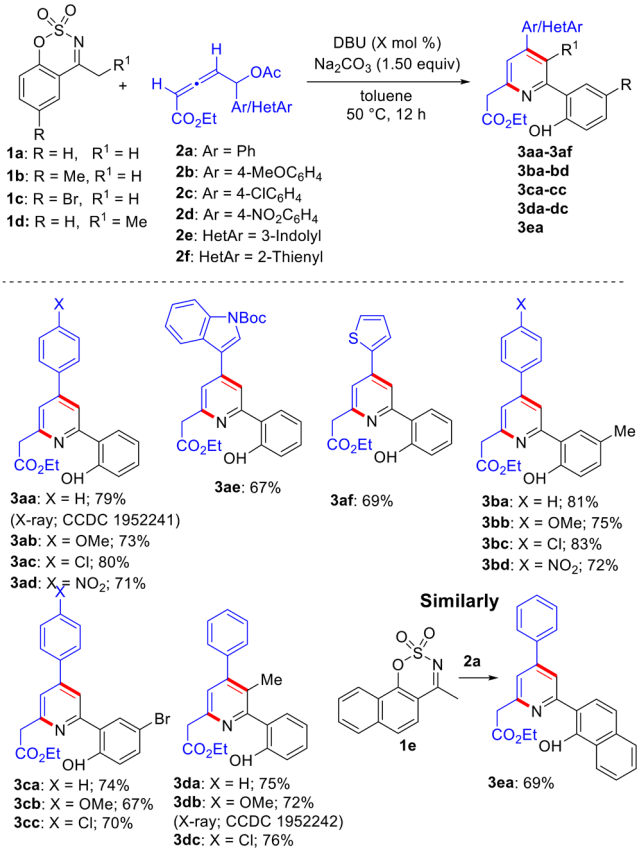
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cleaved product **5aa** was obtained in 73% yield at 80 °C (entry 12); by the exclusion of Na_2CO_3, the yield of this product could be enhanced further (entry 13). Pleasingly, at rt, we obtained sulfomoyloxy carboxylate **4aa** exclusively, with an isolated yield of 73% (entry 14); increasing the reaction time was not required here also (entry 15). In the absence of TPP and Na_2CO_3, we did not observe any reaction even after 24 h.

Coming to the substrate scope, as depicted in Table 2, δ-acetoxy allenoates possessing different functionalities (OMe, Cl, and NO_2) on the phenyl ring furnished the 2-pyridinyl acetates in good to high yields (**3aa**–**3ad**). Heterocyclic allenoates **2e** and **2f** also readily underwent [3 + 3] annulation with **1a** to afford the desired pyridyl acetates **3ae** and **3af** in good yield (67–69%). The allenoate **2b** with the electron-donating group (4-OMe) on the phenyl ring delivered the pyridine cores in slightly higher yield than the allenoate **2d** containing an electron-withdrawing group (NO_2). Delightedly, this protocol was efficient in the reaction of a nucleophilic partner containing electron-poor (−Br; **1c**) as well as electron-rich (Me; **1b**) functionalities with the δ-acetoxy allenoates **2a**–**2d**, affording the corresponding α-pyridyl acetates (**3ba**–**3bd** and **3ca**–**3cc**) in 67–83% yield. In the case of precursor **1d**, we had to use 1 mole equiv of DBU to get the best yield of **3da**–**3dc** (72–76%). It is remarkable that the cyclic sulfamidate imine **1e** with a naphthyl group also enabled us to obtain the corresponding acetate **3ea** in 69% yield under our protocol.

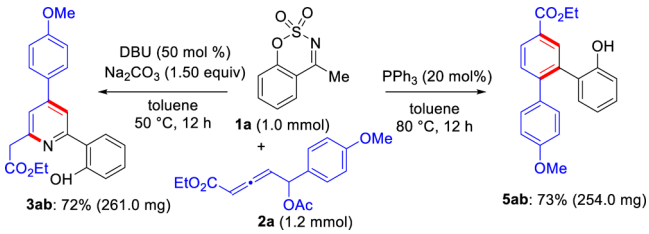
To confirm the feasibility of the present [3 + 3] and [4 + 2] annulations, scale-up experiments were carried out between **1a** (1.0 mmol) and **2b** (1.2 mmol) under the optimized conditions (Scheme 3). The reactions worked well, and we isolated the corresponding products with good yields.

Table 2. Scope of [3 + 3] Annulation in the Reaction of Cyclic Sulfamidate Imines with δ-Acetoxy Allenoates[a]



[a]Reaction conditions: (i) **1a**–**1c** or **1e** (0.20 mmol), **2a**–**2f** (0.24 mmol), DBU (0.10 mmol), and Na_2CO_3 (0.30 mmol) in toluene (2.0 mL) at 50 °C; (ii) **1d** (0.20 mmol), **2a**–**2c** (0.24 mmol), DBU (0.20 mmol), and Na_2CO_3 (0.30 mmol) in toluene (2.0 mL) at 50 °C. Yields given are after isolation.

Scheme 3. Scale-Up Experiments



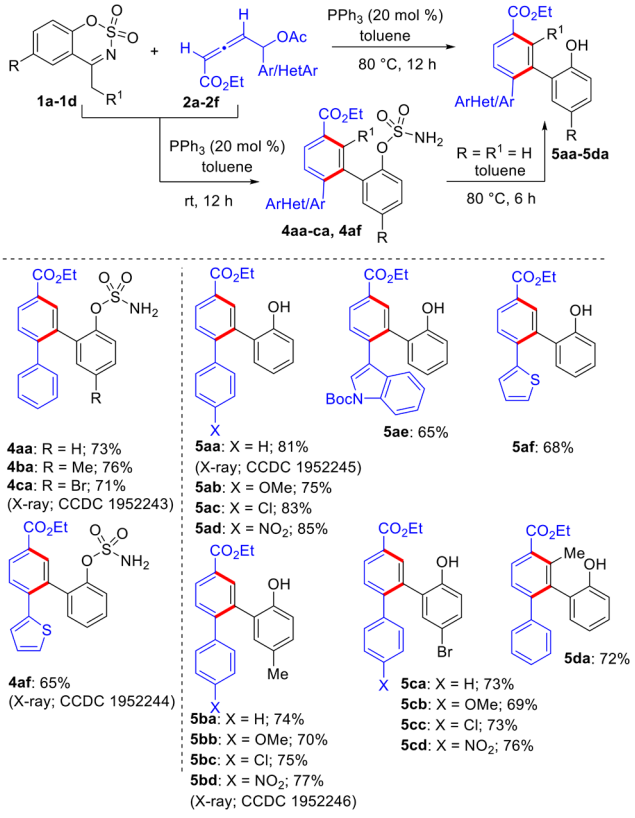
As mentioned above, the [4 + 2] annulation product, terphenyl sulfomoyloxy carboxylate **4aa**, was obtained in good yield by performing the reaction of **1a** with **2a** at rt in the presence of Ph_3P (cf. Table 1, entry 14). By employing the same conditions and using cyclic sulfamidate imines (**1b**–**1c**), we could obtain sulfomoyloxy carboxylates **4ba** and **4ca** in decent yields (Table 3). Satisfyingly, the heterocyclic allenoate **2f** also afforded the annulated product **4af** in good yield (65%).

The phenolic terphenyl **5aa** must have formed by the elimination of $-SO_2(NH_2)$ moiety from **4aa**. This could be proven readily by heating compound **4aa** in toluene that afforded **5aa** quantitatively by S–O bond cleavage. Since such phenolic teraryls are useful scaffolds in organic synthesis,[19] we prepared the analogous of multisubstituted phenolic teraryls **5aa**–**5ad** in one pot in 75–85% yields. Precursors **1b** and **1c** also provided the [4 + 2] annulation products **5ba**–**5bd** and **5ca**–**5cd** in good yields. Even the ethyl substituted oxathiazine-2,2-dioxide **1d** delivered the product **5da** in 72% yield. Heterocyclic allenoates **2e** and **2f** also participated efficiently to afford the products **5ae** and **5af**. All these are depicted in Table 3.

In contrast to the sulfamidate imines **1a**–**1e** that contain an OSO_2 moiety, leading to products **3** with a phenolic −OH, the sulfonyl imines **6a**–**6b** have the sulfur directly bonded to the arene. Hence, in reactions using **6a**–**6b**, the SO_2 moiety has to be eliminated to obtain products similar to **3**. Our initial attempt to treat **6a** with **2a** under the [3 + 3] annulation protocol (cf. Table 1, entry 7) only led to a trace amount (<5%) of **7aa**; at 100 °C, 25% yield was obtained. Finally, we found that 1 mole equiv of DBU provided **7aa** in 71% yield at 100 °C/12 h. As shown in Table 4, **6a**–**6b** showed good reactivity with **2b**–**2c** and produced 2-pyridinyl acetates **7ab**–**7ac** and **7ba**–**7bc** in acceptable yields. Remarkably, the heterocyclic acetoxy allenoates **2e**–**2f** also participated efficiently in this [3 + 3] annulation to produce **7ae**–**7af** in decent yields. In an effort to identify/isolate intermediates for this [3 + 3] annulation, we performed the reaction between **6b** and **2b** using 1.5 equiv of DBU at 60 °C. Pleasingly, after 12 h, we isolated the fused dihydro-pyridine **8bb** in 74% yield whose structure was confirmed by X-ray crystallography. This product **8bb** could be converted to 2-pyridinyl acetate **7bb** (^{1}H NMR/HRMS evidence) in toluene under basic conditions at 100 °C.

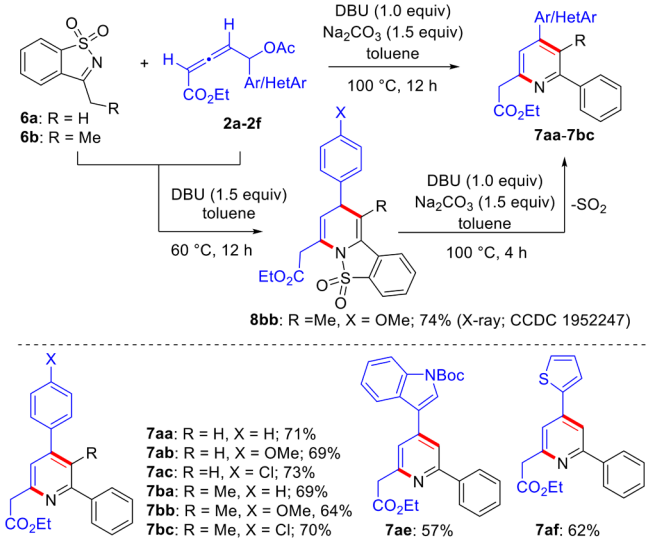
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Table 3. Scope of [4 + 2] Annulation in the Reaction of Cyclic Sulfamidate Imines with δ-Acetoxy Allenoates[a]



[a]Reaction conditions: **1a−1d** (0.20 mmol), **2a−2f** (0.24 mmol), and PPh_3 (0.04 mmol) in toluene (2.0 mL) at rt/12 h for **4** and 80 °C/12 h for **5**. Yields given are after isolation.

Table 4. Scope of [3 + 3] Annulation in the Reaction of Cyclic Sulfonyl Imines with δ-Acetoxy Allenoates[a]

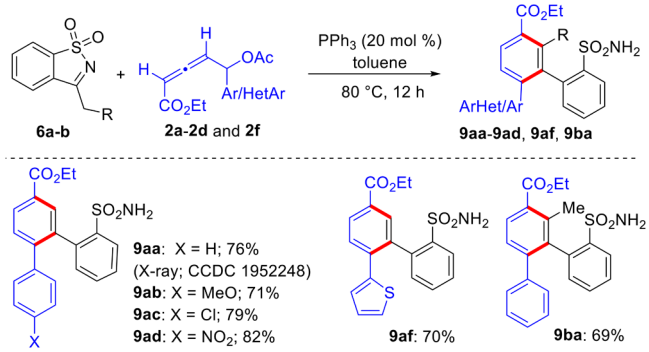


[a]Reaction conditions for one pot reaction to give **7aa−7bc**: **6a−6b** (0.20 mmol), **2a−2f** (0.24 mmol), DBU (0.20 mmol), and Na_2CO_3 (0.30 mmol) in toluene (2.0 mL) at 100 °C (oil bath); conditions for **7bb** via intermediate **8bb**: **6b** (0.20 mmol), **2b** (0.24 mmol), and DBU (0.30 mmol) in toluene (2.0 mL) at 60 °C; then **8bb** (0.10 mmol), DBU (0.10 mmol), and Na_2CO_3 (0.15 mmol) in toluene (1.0 mL) at 100 °C (oil bath). Yields given are after isolation.

Based on this observation, we conclude that the 2-pyridinyl acetates **7** are formed by the elimination of $-SO_2$ group (C−S bond cleavage) from the fused dihydro-pyridine motifs of type **8bb**.

Inspired by the above success with cyclic sulfamidates in [4 + 2] annulation, we tested this protocol with sulfonyl imines **6a−6b** (Table 5). Pleasingly, terphenyls **9aa−9ad**, **9af**, and **9ba** that retain the sulfonamide group are obtained in high to excellent yields. Thus, in contrast to compounds of type **5**, the sulfonyl group is intact here since sulfur is directly attached to the arene ring.

Table 5. Scope of [4 + 2] Annulation in the Reaction of Cyclic Sulfonyl Imines with δ-Acetoxy Allenoates[a]

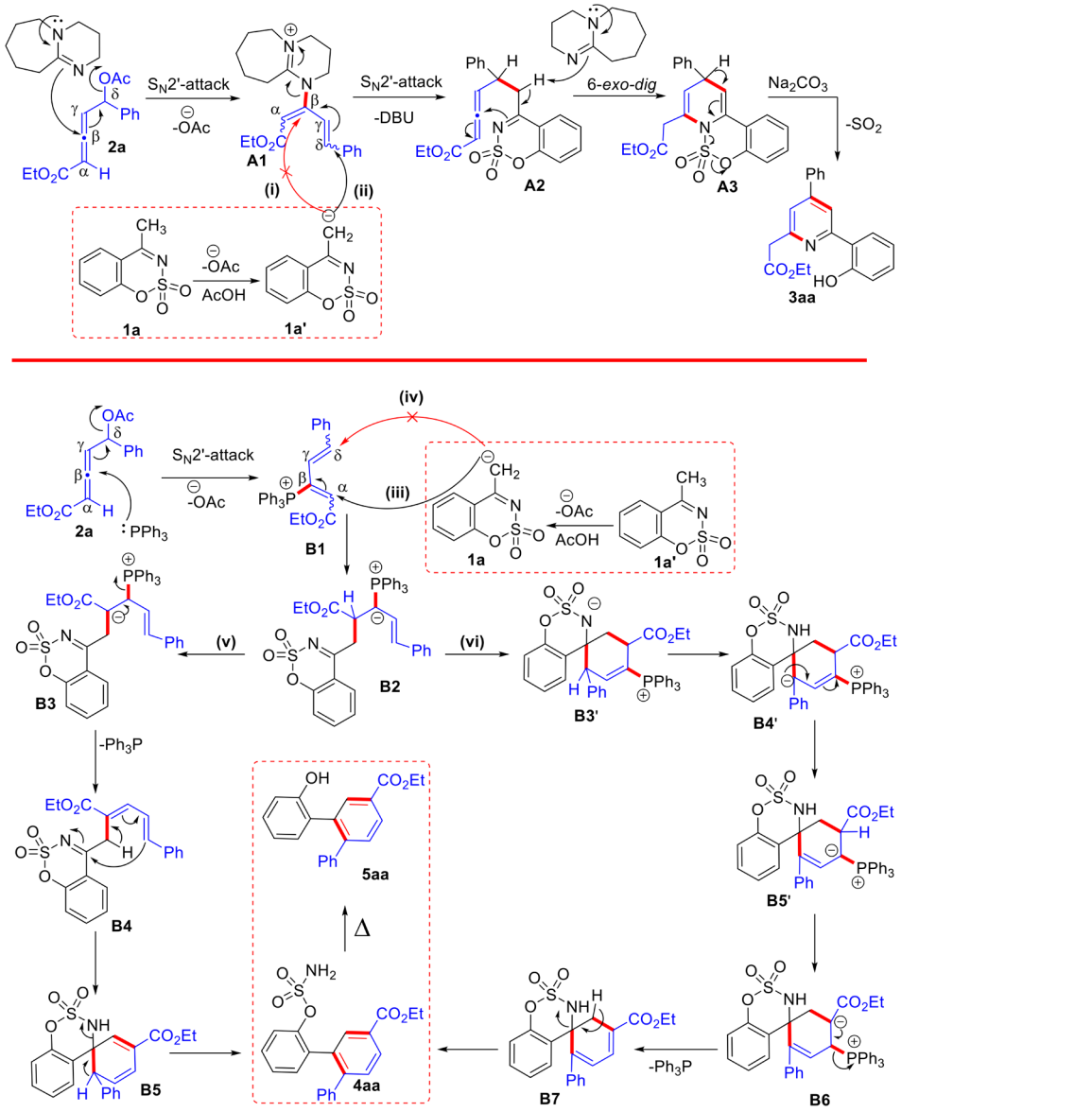


[a]Reaction conditions: **6a−6b** (0.20 mmol), **2a−2d**, and **2f** (0.24 mmol), and PPh_3 (0.04 mmol) in toluene (2.0 mL), at 80 °C. Yields given are after isolation.

Based partly on the previous literature,[20] plausible pathways for the above [3 + 3] and [4 + 2] annulations are shown in Scheme 4. We could observe peaks assignable to intermediates **A1**, **A2**/**A3**, **B1**, and **B2** by checking crude reaction mixtures using ESI-HRMS (see the Supporting Information). Initially, the δ-acetoxy allenoate **2a** undergoes S_N2' attack with Lewis base (DBU[20d] or TPP) to deliver the electrophilic intermediate **A1** or **B1** by the elimination of acetate anion. In the [3 + 3] annulation, **A1** is likely to be less electrophilic (at β-carbon), rendering Michael addition (route i) less favorable than DBU elimination. Hence, the cationic intermediate **A1** undergoes a second S_N2' attack (route ii) rapidly with carbanion **1a′** generated by deprotonation of **1a** to afford intermediate **A2**. Subsequently, ring closure of **A2** via 6-*exo*-*dig* cyclization leading to **A3** followed by SO_2 elimination/aromatization offers 2-pyridinyl acetate **3aa**. The likely role of Na_2CO_3 in this reaction is to scavenge the acetic acid formed and to facilitate SO_2 elimination.[21] In the [4 + 2] annulation, the α-carbon in the intermediate **B1** is more electrophilic. The driving force for Michael addition is ylide formation with **1a′** to give **B2** via route iii (more favorable than route iv). This is followed by 1,2-proton shift to afford **B3** that undergoes TPP elimination, leading to the diene **B4** via route v. Then, **B4** undergoes intramolecular vinylogous Mannich coupling[22] followed by C−N bond cleavage/aromatization, affording sulfamoyloxy-terphenyl-carboxylate **4aa**. Finally, **4aa** produces **5aa** by the cleavage of O−S bond upon heating. Another pathway, which involves more number of steps, is also possible for the phosphine catalysis. Thus, ylide **B2** may involve in Mannich coupling via route vi to afford **B3′** that undergoes 1,3-proton shift, leading to **B4′**. Then, **B4′** undergoes isomerization followed by 1,2-proton shift and TPP elimi-

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J. Org. Chem. 2020, 85, 4130−4144

Scheme 4. Probable Reaction Pathways for the [3 + 3] and [4 + 2] Annulations



nation, affording the spirocyclic diene **B7**. This, in turn, will give the products **4aa** and **5aa**.

CONCLUSIONS

In this study, we have described new Lewis base-switched [3 + 3] and [4 + 2] annulation protocols involving cyclic *N*-sulfonyl imines and *δ*-acetoxy allenoates generating functional pyridinyl acetates and terphenyls, respectively, under metal-free conditions. While in amine-mediated [3 + 3] annulation, cyclic *N*-sulfonyl imines act as C/N donors and afford 2-pyridyl acetates via double S$_N$2′−S$_N$2′ attack followed by aza-Michael addition and SO_2 elimination/aromatization, in TPP-catalyzed [4 + 2] annulation, cyclic *N*-sulfonyl imines act as C/C donors and deliver terphenyl scaffolds via a S$_N$2′ attack followed by Michael addition, intramolecular direct vinylogous Mannich coupling, and C−N bond cleavage/aromatization.

EXPERIMENTAL SECTION

General Methods. All the reactions were carried out either under an inert atmosphere or air. All required chemicals were procured from Aldrich or local manufacturers and used as such without further purification, unless noted. Melting points have been recorded on an MR-visual melting range apparatus and are uncorrected. The ^{1}H and ^{13}C NMR spectra were recorded using 5 mm tubes on a 400 and 500 MHz NMR spectrometer [field strengths: 400/100 or 500/125 MHz, respectively] in $CDCl_3$ solution (unless specified otherwise) with shifts referenced to TMS (^{1}H, ^{13}C = 0). Data for ^{1}H and ^{13}C NMR are reported as a chemical shift (*δ*, ppm). All *J* values are in Hz. The infrared spectra were recorded using the attenuated total reflection technique on a Fourier-transform infrared spectrophotometer. The mass spectra were recorded using HRMS (ESI-TOF analyzer) equipment. Crystallographic data were collected on a X-ray diffractometer system using Mo-K$_\alpha$ radiation (λ = 0.71073 Å) radiation. Structures were solved and refined using standard methods. Thin-layer chromatography (TLC) was performed on silica/alumina plates, and components were visualized under iodine/UV light at 254 nm. Column chromatography was performed on silica gel (100−200 mesh); for the column elution process, hexane-EtOAc mixture was used as the eluent, unless otherwise stated.

Cyclic sulfamidate imines **1a−1e**,[23] cyclic sulfonyl imines **1f−1g**,[24] and *δ*-acetoxy allenoates **2a−2f**[25] were synthesized using literature

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J. Org. Chem. 2020, 85, 4130−4144

procedures. The solvent of crystallization for all the products was ethyl acetate-hexane (ca. 4:1 v/v).

(i) Procedure for the Synthesis of δ-Acetoxy Allenoates 2d and 2e. *Ethyl 5-Acetoxy-5-(4-nitrophenyl)penta-2,3-dienoate (**2d**).* To a solution of ethyl 5-hydroxy-5-(4-nitrophenyl)penta-2,3-dienoate (2.63 g, 10 mmol) in dry dichloromethane (20 mL) at 0 °C was added triethylamine (2.80 mL, 20.0 mmol, 2.0 equiv), and then the mixture stirred for 30 min at the same temperature. After that, acetyl chloride (0.85 mL, 12.0 mmol, 1.2 equiv) was slowly added into the mixture over 5 min, and the contents were stirred for 40 min at 0 °C. After completion of the reaction (TLC), the aqueous layer was extracted with dichloromethane (3 × 25 mL). Then, the combined organic layer was washed with brine (2 × 20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (10:90) as the eluent. This afforded **2d** as a diasteromeric mixture (~1:0.9 ratio; in the ^{1}H NMR for minor isomer, corresponding protons are indicated by the prime (′) symbol). Yield, 1.70 g (55%); mp 75−77 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.22 (d, *J* = 8.5 Hz, 2H + 2H′), 7.61 (d, *J* = 9.0 Hz, 2H + 2H′), 6.46−6.42 (m, 1H + 1H′), 5.91 (t, *J* = 6.0 Hz, 1H′), 5.85 (t, *J* = 6.5 Hz, 1H), 5.75 (dd, *J* = 2.0, 6.0 Hz, 1H), 5.72 (dd, *J* = 2.5, ~6.5 Hz, 1H′), 4.22−4.16 (m, 2H + 2H′), 2.16 (s, 3H), 2.13 (s, 3H′), 1.30−1.27 (m, 3H + 3H′); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ (major isomer) 212.6, 169.6, 164.7, 148.0, 145.1, 127.8, 123.9, 96.0, 92.0, 71.6, 61.4, 21.0, 14.3; (minor isomer) 212.3, 169.6, 164.7, 148.0, 144.8, 128.2, 123.9, 96.2, 91.5, 71.0, 61.4, 21.1, 14.3; IR (neat): ν_{max} 2980, 1968, 1708, 1516, 1446, 1369, 1216, 1105, 1026, 744 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{15}H_{15}NNaO_6$, 328.0792; found, 328.0784.

*tert-Butyl 3-(1-Acetoxy-5-ethoxy-5-oxopenta-2,3-dien-1-yl)-1H-indole-1-carboxylate (**2e**).* Following the procedure for **2d**, the reaction of *tert*-butyl 3-(5-ethoxy-1-hydroxy-5-oxopenta-2,3-dien-1-yl)-1*H*-indole-1-carboxylate (3.57 g, 10 mmol) with acetyl chloride (0.85 mL, 12 mmol) and triethylamine (2.80 mL, 20 mmol) afforded **2e** as a gummy liquid and a diastereomeric mixture (*dr* = 1:0.9; in the ^{1}H NMR for minor isomer corresponding protons are indicated by the prime (′) symbol). Yield, 1.91 g (48%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.17−8.16 (m, 1H + 1H′), 7.76−7.67 (m, 2H + 2H′), 7.38−7.34 (m, 2H′), 7.29−7.25 (m, 2H), 6.72−6.68 (m, 1H + 1H′), 6.04−6.00 (m, 1H + 1H′), 5.80 (dd, *J* = 6.0, 2.5 Hz, 1H), 5.77 (dd, *J* = 6.0, 2.5 Hz, 1H′), 4.26−4.17 (m, 2H + 2H′), 2.13 (s, 3H), 2.12 (s, 3H′), 1.70 (s, 9H + 9H′), 1.31 (t, *J* = 7.0 Hz, 3H′), 1.27 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ (major isomer) 212.4, 170.0, 165.0, 149.6, 128.2, 125.1, 125.0, 124.6, 123.0, 120.0, 118.0, 115.5, 96.0, 91.7, 84.3, 65.6, 61.2, 28.3, 21.1, 14.3; (minor isomer) 212.0, 170.0, 165.1, 149.6, 128.2, 125.1, 125.0, 124.6, 122.9, 120.0, 117.7, 115.5, 95.8, 91.4, 84.2, 66.2, 61.2, 28.3, 21.1, 14.3; IR (neat): ν_{max} 2979, 1967, 1717, 1452, 1367, 1218, 1152, 1084, 746 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{22}H_{25}NNaO_6$, 422.1574; found, 422.1578.

(ii) General Procedure for the Synthesis of Compounds 3aa−3ea. A Schlenk tube was charged with cyclic sulfamidate imine **1** (0.20 mmol), δ-acetoxy allenoate **2** (0.24 mmol), Na_2CO_3 (0.30 mmol), and 1.5 mL of toluene. Subsequently, DBU (0.10 mmol) in 0.5 mL of toluene was added, the mixture was stirred at 50 °C (oil bath) for the stipulated time, and the reaction progress was monitored using TLC. After completion of the reaction, the mixture was quenched by adding water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then the combined organic layer was washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (10:90) as the eluent.

*Ethyl 2-(6-(2-Hydroxyphenyl)-4-phenylpyridin-2-yl)acetate (**3aa**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3aa** as a white solid. Yield, 53.0 mg (79%); mp 100−102 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 13.98 (s, 1H), 8.00 (d, *J* = 1.0 Hz, 1H), 7.88 (dd, *J* = 8.0, 1.5 Hz, 1H), 7.70−7.67 (m, 2H), 7.54−7.43 (m, 3H), 7.43 (d, *J* = 1.5 Hz, 1H), 7.34−7.31 (m, 1H), 7.04 (dd, *J* = 8.0, 1.5 Hz, 1H), 6.95−6.91 (m, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.93 (s, 2H), 1.31 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.0, 160.0, 158.2, 151.8, 151.2, 138.2, 131.7, 129.6, 129.3, 127.3, 126.5, 120.1, 119.1, 119.0, 118.8, 116.0, 61.6, 43.4, 14.3; IR (neat): ν_{max} 3438 (w), 3059, 2978, 1727, 1607, 1554, 1416, 1322, 1191, 1025, 746, 678 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{21}H_{20}NO_3$, 334.1438; found, 334.1439.

*Ethyl 2-(6-(2-Hydroxyphenyl)-4-(4-methoxyphenyl)pyridin-2-yl)-acetate (**3ab**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ab** as a white solid. Yield, 53.0 mg (73%); mp 99−101 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 14.03 (s, 1H), 7.96 (d, *J* = 0.5 Hz, 1H), 7.88 (dd, *J* = 8.0, 1.5 Hz, 1H), 7.66−7.64 (m, 2H), 7.39 (d, *J* = 1.0 Hz, 1H), 7.33−7.29 (m, 1H), 7.04−7.02 (m, 3H), 6.94−6.91 (m, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.91 (s, 2H), 3.88 (s, 3H), 1.31 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.0, 161.0, 160.0, 158.0, 151.7, 150.6, 131.6, 130.4, 128.5, 126.5, 119.5, 119.2, 118.9, 118.7, 115.3, 114.7, 61.5, 55.5, 43.4, 14.2; IR (neat): ν_{max} 3444 (w), 2928, 2835, 1723, 1605, 1508, 1410, 1325, 1173, 1033, 767, 655 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{22}NO_4$, 364.1543; found, 364.1545.

*Ethyl 2-(4-(4-Chlorophenyl)-6-(2-hydroxyphenyl)pyridin-2-yl)-acetate (**3ac**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ac** as a white solid. Yield, 59.0 mg (80%); mp 149−151 °C; ^{1}H NMR (400 MHz, $CDCl_3$): δ 13.87 (s, 1H), 7.93 (d, *J* = 1.0 Hz, 1H), 7.85 (dd, *J* = 8.0, 1.5 Hz, 1H), 7.62−7.60 (m, 2H), 7.50−7.47 (m, 2H), 7.38 (d, *J* = 1.5 Hz, 1H), 7.34−7.30 (m, 1H), 7.04 (dd, *J* = 8.0, 1.0 Hz, 1H), 6.94−6.91 (m, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.92 (s, 2H), 1.31 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.9, 160.0, 158.3, 152.1, 149.9, 136.7, 135.9, 131.8, 129.6, 128.6, 126.5, 119.8, 119.0, 118.8, 115.7, 61.6, 43.4, 14.3; IR (neat): ν_{max} 3441 (w), 2974, 2942, 1726, 1651, 1496, 1186, 1026, 794, 648 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{21}H_{19}{}^{35}ClNO_3$ and $C_{21}H_{19}{}^{37}ClNO_3$, 368.1048 and 370.1018; found, 368.1049 and 370.1019.

*Ethyl 2-(6-(2-Hydroxyphenyl)-4-(4-nitrophenyl)pyridin-2-yl)-acetate (**3ad**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2d** (73.2 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ad** as a light brown solid. Yield, 54.0 mg (71%); mp 176−178 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 13.67 (s, 1H), 8.40 (d, *J* = 8.5 Hz, 2H), 8.00 (s, 1H), 7.87−7.83 (m, 3H), 7.45 (s, 1H), 7.36−7.33 (m, 1H), 7.05 (d, *J* = 8.5 Hz, 1H), 6.96−6.93 (m, 1H), 4.25 (q, *J* = 7.0 Hz, 2H), 3.96 (s, 2H), 1.32 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 169.8, 160.0, 158.7, 152.5, 148.8, 148.5, 144.6, 132.2, 128.4, 126.5, 124.6, 120.1, 119.2, 118.9, 118.8, 116.2, 61.7, 43.4, 14.3; IR (neat): ν_{max} 3444 (w), 2986, 2919, 2846, 1729, 1610, 1594, 1444, 1187, 1022, 749, 688 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{21}H_{19}N_2O_5$, 379.1288; found, 379.1289.

*tert-Butyl 3-(2-(2-Ethoxy-2-oxoethyl)-6-(2-hydroxyphenyl)-pyridin-4-yl)-1H-indole-1-carboxylate (**3ae**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2e** (95.8 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ae** as a white solid. Yield, 63.2 mg (67%); mp 131−133 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 13.96 (s, 1H), 8.27 (d, *J* = 8.0 Hz, 1H), 8.07 (s, 1H), 7.94 (s, 1H), 7.88−7.86 (m, 2H), 7.49 (s, 1H), 7.45−7.42 (m, 1H), 7.39−7.32 (m, 2H), 7.05 (d, *J* = 8.0 Hz, 1H), 6.96−6.93 (m, 1H), 4.26 (q, *J* = 7.0 Hz, 2H), 3.94 (s, 2H), 1.73 (s, 9H), 1.32 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.0, 160.1, 158.3, 152.0, 149.5, 144.5, 136.2, 131.7, 128.0, 126.5, 125.4, 124.9, 123.7, 120.5, 119.7, 119.5, 119.2, 119.0, 118.8, 116.4, 115.9, 84.8, 61.6, 43.5, 28.4, 14.3; IR (neat): ν_{max} 3059, 2974, 2923, 2856, 1730,

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J. Org. Chem. 2020, 85, 4130−4144

1608, 1453, 1369, 1152, 1061, 761, 645 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{28}H_{29}N_2O_5$, 473.2071; found, 473.2074.

*Ethyl 2-(6-(2-Hydroxyphenyl)-4-(thiophen-2-yl)pyridin-2-yl)-acetate (**3af**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2f** (63.8 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3af** as a white solid. Yield, 46.7 mg (69%); mp 144−146 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 13.85 (s, 1H), 7.96 (d, *J* = 1.0 Hz, 1H), 7.85 (dd, *J* = 8.0, 1.5 Hz, 1H), 7.59−7.58 (m, 1H), 7.47 (dd, *J* = 5.0, 1.0 Hz, 1H), 7.41−7.40 (m, 1H), 7.34−7.30 (m, 1H), 7.17 (dd, *J* = 5.0, 3.5 Hz, 1H), 7.03 (dd, *J* = 8.0, 1.0 Hz, 1H), 6.95−6.92 (m, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.89 (s, 2H), 1.31 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.9, 160.1, 158.4, 152.1, 144.0, 141.0, 131.8, 128.7, 127.9, 126.5, 126.2, 119.0$_2$, 118.9$_9$, 118.8, 118.2, 114.1, 61.6, 43.4, 14.3; IR (neat): ν_{max} 3430 (w), 3069, 2983, 2904, 1726, 1607, 1555, 1432, 1201, 1178, 755, 621 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{19}H_{18}NO_3S$, 340.1002; found, 340.1006.

*Ethyl 2-(6-(2-Hydroxyphenyl)-5-methyl-4-phenylpyridin-2-yl)-acetate (**3ba**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ba** as a white solid. Yield, 56.2 mg (81%); mp 101−103 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 13.73 (s, 1H), 7.97 (s, 1H), 7.69 (d, *J* = 7.0 Hz, 2H), 7.65 (s, 1H), 7.54−7.47 (m, 3H), 7.41 (s, 1H), 7.13 (d, *J* = 8.0 Hz, 1H), 6.95 (d, *J* = 8.5 Hz, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.92 (s, 2H), 2.35 (s, 3H), 1.31 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.0, 158.2, 157.7, 151.8, 151.1, 138.3, 132.5, 129.5, 129.3, 127.9, 127.3, 126.6, 120.0, 118.7, 118.5, 115.9, 61.5, 43.4, 20.9, 14.3; IR (neat): ν_{max} 3447 (w), 3034, 2983, 2903, 1722, 1601, 1545, 1372, 1215, 1023, 761, 691 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{22}H_{22}NO_3$, 348.1594; found, 348.1593.

*Ethyl 2-(6-(2-Hydroxy-5-methylphenyl)-4-(4-methoxyphenyl)-pyridin-2-yl)acetate (**3bb**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.20 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3bb** as a white solid. Yield, 56.5 mg (75%); mp 113−115 °C; ^1H NMR (400 MHz, $CDCl_3$): δ 13.77 (s, 1H), 7.94 (s, 1H), 7.67−7.65 (m, 3H), 7.37 (s, 1H), 7.12 (dd, *J* = 8.4, 2.0 Hz, 1H), 7.05−7.02 (m, 2H), 6.94 (d, *J* = 8.0 Hz, 1H), 4.24 (q, *J* = 7.2 Hz, 2H), 3.90 (s, 2H), 3.88 (s, 3H), 2.35 (s, 3H), 1.30 (t, *J* = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.1, 160.9, 158.1, 157.8, 151.7, 150.5, 132.4, 130.5, 128.6, 127.8, 126.6, 119.4, 118.8, 118.5, 115.2, 114.7, 61.5, 55.6, 43.5, 20.9, 14.3; IR (neat): ν_{max} 3434 (w), 2996, 2913, 2839, 1722, 1604, 1515, 1496, 1180, 1030, 766, 671 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{23}H_{24}NO_4$, 378.1700; found, 378.1703.

*Ethyl 2-(4-(4-Chlorophenyl)-6-(2-hydroxy-5-methylphenyl)-pyridin-2-yl)acetate (**3bc**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3bc** as a white solid. Yield, 63.2 mg (83%); mp 130−132 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 13.60 (s, 1H), 7.92 (s, 1H), 7.63−7.61 (m, 3H), 7.49 (d, *J* = 8.5 Hz, 2H), 7.37 (s, 1H), 7.14−7.12 (m, 1H), 6.94 (d, *J* = 8.5 Hz, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.91 (s, 2H), 2.35 (s, 3H), 1.30 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 169.9, 158.4, 157.7, 152.1, 149.8, 136.7, 135.8, 132.7, 129.5, 128.6, 127.9, 126.6, 119.7, 118.5$_9$, 118.5$_7$, 115.7, 61.6, 43.4, 20.9, 14.3; IR (neat): ν_{max} 3426 (w), 2978, 2867, 1723, 1606, 1497, 1180, 1029, 772, 664 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{22}H_{21}{}^{35}ClNO_3$ and $C_{22}H_{21}{}^{37}ClNO_3$, 382.1204 and 384.1175; found, 382.1205 and 384.1178.

*Ethyl 2-(6-(2-Hydroxy-5-methylphenyl)-4-(4-nitrophenyl)pyridin-2-yl)acetate (**3bd**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2d** (73.2 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3bd** as a pale yellow solid. Yield, 56.4 mg (72%); mp 179−181 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 13.45 (s, 1H), 8.40−8.38 (m, 2H), 7.98 (d, *J* = 1.0 Hz, 1H), 7.86−7.85 (m, 2H), 7.64 (d, *J* = 1.5 Hz, 1H), 7.44 (d, *J* = 1.0 Hz, 1H), 7.16 (dd, *J* = 8.0, 1.5 Hz, 1H), 6.95 (d, *J* = 8.0 Hz, 1H), 4.25 (q, *J* = 7.0 Hz, 2H), 3.96 (s, 2H), 2.36 (s, 3H), 1.31 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.8, 158.8, 157.8, 152.6, 148.8, 148.6, 144.7, 133.1, 128.5, 128.2, 126.6, 124.6, 119.9, 118.7, 118.4, 116.2, 61.7, 43.4, 20.9, 14.3; IR (neat): ν_{max} 3428 (w), 2987, 2920, 2849, 1723, 1609, 1500, 1193, 791, 688 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{22}H_{21}N_2O_5$, 393.1445; found, 393.1446.

*Ethyl 2-(6-(5-Bromo-2-hydroxyphenyl)-4-phenylpyridin-2-yl)-acetate (**3ca**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ca** as a white solid. Yield, 61.0 mg (74%); mp 137−139 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 14.02 (s, 1H), 7.95 (s, 1H), 7.91 (s, 1H), 7.68 (d, *J* = 7.0 Hz, 2H), 7.54−7.48 (m, 3H), 7.46 (s, 1H), 7.38 (dd, *J* = 8.5, 2.0 Hz, 1H), 6.92 (d, *J* = 9.0 Hz, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.93 (s, 2H), 1.31 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.8, 159.1, 156.7, 152.0, 151.5, 137.8, 134.2, 129.7, 129.4, 128.9, 127.3, 120.7, 120.6, 116.0, 110.7, 61.6, 43.3, 14.3; IR (neat): ν_{max} 3434 (w), 3061, 2972, 2930, 1729, 1603, 1543, 1475, 1153, 1027, 760, 683 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{21}H_{19}{}^{79}BrNO_3$ and $C_{21}H_{19}{}^{81}BrNO_3$, 412.0543 and 414.0522; found, 412.0539 and 414.0523.

*Ethyl 2-(6-(5-Bromo-2-hydroxyphenyl)-4-(4-methoxyphenyl)-pyridin-2-yl)acetate (**3cb**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3cb** as a white solid. Yield, 59.1 mg (67%); mp 149−151 °C; ^1H NMR (400 MHz, $CDCl_3$): δ 14.05 (s, 1H), 7.95 (d, *J* = 2.4 Hz, 1H), 7.88 (d, *J* = 1.2 Hz, 1H), 7.67−7.63 (m, 2H), 7.42 (d, *J* = 1.2 Hz, 1H), 7.37 (dd, *J* = 8.8, 2.4 Hz, 1H), 7.06−7.02 (m, 2H), 6.92 (d, *J* = 8.8 Hz, 1H), 4.24 (q, *J* = 7.2 Hz, 2H), 3.91 (s, 2H), 3.89 (s, 3H), 1.30 (t, *J* = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.9, 161.1, 159.1, 156.6, 151.8, 150.9, 134.1, 130.0, 128.9, 128.6, 120.8, 120.6, 120.1, 115.3, 114.8, 110.7, 61.6, 55.6, 43.3, 14.3; IR (neat): ν_{max} 3444 (w), 3093, 2977, 2931, 1726, 1602, 1516, 1389, 1172, 1028, 769, 672 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{22}H_{21}{}^{79}BrNO_4$ and $C_{22}H_{21}{}^{81}BrNO_4$, 442.0648 and 444.0628; found, 442.0648 and 444.0631.

*Ethyl 2-(6-(5-Bromo-2-hydroxyphenyl)-4-(4-chlorophenyl)-pyridin-2-yl)acetate (**3cc**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol) and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3cc** as a white solid. Yield, 62.3 mg (70%); mp 145−147 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 13.88 (s, 1H), 7.93 (d, *J* = 2.0 Hz, 1H), 7.87 (d, *J* = 1.0 Hz, 1H), 7.63−7.61 (m, 2H), 7.52−7.49 (m, 2H), 7.42 (d, *J* = 1.0 Hz, 1H), 7.38 (dd, *J* = 8.5, 2.0 Hz, 1H), 6.92 (d, *J* = 8.5 Hz, 1H), 4.24 (q, *J* = 7.0 Hz, 2H), 3.92 (s, 2H), 1.31 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 169.8, 159.1, 157.0, 152.3, 150.3, 136.3, 136.1, 134.4, 129.7, 129.0, 128.7, 120.7$_3$, 120.6$_6$, 120.5, 115.8, 110.8, 61.7, 43.3, 14.3; IR (neat): ν_{max} 3429 (w), 2986, 2900, 1722, 1607, 1472, 1290, 1165, 1027, 823, 767, 663 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + Na]^+$ calcd for $C_{21}H_{17}{}^{35}Cl^{79}BrNNaO_3$, $C_{21}H_{17}{}^{37}Cl^{79}BrNNaO_3$/$C_{21}H_{17}{}^{35}Cl^{81}BrNNaO_3$, and $C_{21}H_{17}{}^{37}Cl^{81}BrNNaO_3$, 467.9973, 469.9952, and 471.9948; found, 467.9974, 469.9955, and 471.9930.

*Ethyl 2-(6-(2-Hydroxyphenyl)-5-methyl-4-phenylpyridin-2-yl)-acetate (**3da**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1d** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (30.4 mg, 0.20 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3da** as a gummy liquid. Yield, 52.0 mg (75%); ^1H NMR (500 MHz, $CDCl_3$): δ 11.08 (s, 1H), 7.52 (dd, *J* = 7.5, 1.5 Hz, 1H), 7.49−7.46 (m, 2H), 7.44−7.41 (m, 1H), 7.38−7.36 (m, 2H), 7.30−7.27 (m, 1H), 7.13 (s, 1H), 7.09 (dd, *J* = 8.0, 1.0 Hz, 1H), 6.94−6.90 (m, 1H), 4.23 (q, *J* = 7.0 Hz, 2H), 3.87 (s, 2H), 2.34 (s, 3H), 1.30 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125

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J. Org. Chem. 2020, 85, 4130−4144

MHz, $CDCl_3$): δ 170.4, 157.2, 157.1, 154.2, 149.5, 139.8, 130.6, 130.4, 128.8, 128.7, 128.4, 128.2, 123.3, 122.8, 118.7, 118.2, 61.5, 42.8, 19.3, 14.3; IR (neat): ν_{max} 3423 (w), 3063, 2983, 2935, 2853, 1731, 1595, 1420, 1151, 1032, 733, 700 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{22}H_{22}NO_3$, 348.1594; found, 348.1597.

*Ethyl 2-(6-(2-Hydroxyphenyl)-4-(4-methoxyphenyl)-5-methylpyridin-2-yl)acetate (**3db**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1d** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol), DBU (30.4 mg, 0.20 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3db** as a white solid. Yield, 54.3 mg (72%); mp 114−116 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 11.17 (s, 1H), 7.52 (d, *J* = 7.5 Hz, 1H), 7.32−7.27 (m, 3H), 7.12 (s, 1H), 7.09 (d, *J* = 8.0 Hz, 1H), 7.00 (d, *J* = 8.5 Hz, 2H), 6.93−6.90 (m, 1H), 4.23 (q, *J* = 7.0 Hz, 2H), 3.87−3.86 (m, 5H), 2.36 (s, 3H), 1.30 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.5, 159.8, 157.2$_5$, 157.1$_8$, 153.9, 149.4, 132.1, 130.5$_0$, 130.4$_5$, 130.2, 128.2, 123.3, 122.8, 118.6, 118.1, 114.1, 61.5, 55.5, 42.8, 19.5, 14.3; IR (neat): ν_{max} 3441 (w), 2958, 2838, 1727, 1609, 1513, 1425, 1173, 1029, 836, 762, 654 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{23}H_{24}NO_4$, 378.1700; found, 378.1702.

*Ethyl 2-(4-(4-Chlorophenyl)-6-(2-hydroxyphenyl)-5-methylpyridin-2-yl)acetate (**3dc**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1d** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), DBU (30.4 mg, 0.20 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3dc** as a pale yellow solid. Yield, 58.0 mg (76%); mp 81−83 °C; ^1H NMR (400 MHz, $CDCl_3$): δ 11.03 (s, 1H), 7.50 (dd, *J* = 7.6, 1.6 Hz, 1H), 7.47−7.44 (m, 2H), 7.33−7.27 (m, 3H), 7.10−7.08 (m, 2H), 6.92 (td, *J* = 7.6, 1.2 Hz, 1H), 4.23 (q, *J* = 7.2 Hz, 2H), 3.87 (s, 2H), 2.33 (s, 3H), 1.30 (t, *J* = 7.2 Hz, 3H); $^{13}C\{^1H\}$ NMR (100 MHz, $CDCl_3$): δ 170.4, 157.3, 157.2, 152.9, 149.7, 138.1, 134.6, 130.7, 130.4, 130.2, 128.9, 128.0, 123.0, 122.6, 118.7, 118.2, 61.5, 42.7, 19.3, 14.3; IR (neat): ν_{max} 3390 (w), 2979, 2932, 1731, 1595, 1491, 1151, 1027, 754, 645 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{22}H_{21}{}^{35}ClNO_3$ and $C_{22}H_{21}{}^{37}ClNO_3$, 382.1204 and 384.1175; found, 382.1206 and 384.1174.

*Ethyl 2-(6-(1-Hydroxynaphthalen-2-yl)-4-phenylpyridin-2-yl)-acetate (**3ea**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1e** (49.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (15.2 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ea** as a light brown solid. Yield, 53.0 mg (69%); mp 127−129 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 15.45 (s, 1H), 8.51−8.49 (m, 1H), 8.06 (m, 1H), 7.93 (d, *J* = 8.5 Hz, 1H), 7.78−7.77 (m, 1H), 7.73−7.71 (m, 2H), 7.56−7.48 (m, 5H), 7.42 (d, *J* = 2.0 Hz, 1H), 7.37 (d, *J* = 8.5 Hz, 1H), 4.28 (q, *J* = 7.0 Hz, 2H), 3.98 (s, 2H), 1.35 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 170.0, 158.6, 157.7, 151.6, 151.3, 138.4, 135.4, 129.6, 129.3, 127.8, 127.4, 127.3, 126.6, 125.4, 123.9, 123.0, 119.6, 118.3, 115.9, 111.7, 61.6, 43.5, 14.3; IR (neat): ν_{max} 3447 (w), 3066, 2983, 2932, 1727, 1605, 1544, 1328, 1187, 1023, 799, 697 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + H]^+$ calcd for $C_{25}H_{22}NO_3$, 384.1594; found, 384.1595.

(iii) General Procedure for the Synthesis of Compounds 4aa−4ca. A Schlenk tube was charged with cyclic sulfamidate imine **1** (0.20 mmol), triphenylphosphine (0.04 mmol), and toluene (1.0 mL). Subsequently, δ-acetoxy allenoate **2** (0.24 mmol) in toluene (1.0 mL) was added gradually over 30 min, the mixture was stirred at rt (25 °C) for the stipulated time, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by adding water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then, the combined organic layer was washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (30:70) as the eluent.

*Ethyl 2″-(Sulfamoyloxy)-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**4aa**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **4aa** as a white solid. Yield, 57.9 mg (73%); mp 61−63 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.09−8.07 (m, 2H), 7.52 (d, *J* = 8.5 Hz, 1H), 7.36−7.34 (m, 1H), 7.31−7.28 (m, 1H), 7.27−7.20 (m, 5H), 7.18−7.16 (m, 2H), 4.65 (s, 2H), 4.36 (q, *J* = 7.0 Hz, 2H), 1.39 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.7, 147.6, 146.1, 140.4, 135.6, 133.5, 132.5$_4$, 132.4$_8$, 130.7, 129.4, 129.3$_9$, 129.3$_6$, 128.2, 127.5, 126.7, 121.3, 61.5, 14.4; IR (neat): ν_{max} 3390, 3255, 3061, 2982, 2847, 1698, 1601, 1444, 1370, 1241, 1168, 1049, 756, 700 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + NH_4]^+$ calcd for $C_{21}H_{23}N_2O_5S$, 415.1322; found, 415.1329.

*Ethyl 5″-Methyl-2″-(sulfamoyloxy)-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**4ba**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **4ba** as a white solid. Yield, 62.5 mg (76%); mp 135−137 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.10 (dd, *J* = 8.0, 1.8 Hz, 1H), 8.09−8.07 (m, 1H), 7.53 (d, *J* = 8.0 Hz, 1H), 7.25−7.21 (m, 4H), 7.20−7.18 (m, 2H), 7.12−7.11 (m, 2H), 4.40 (q, *J* = 7.0 Hz, 2H), 4.31 (s, 2H), 2.32 (s, 3H), 1.41(t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.6, 146.0, 145.5, 140.5, 136.7, 135.9, 133.2, 132.9, 132.4, 130.6, 130.0, 129.5$_5$, 129.4$_7$, 129.3$_8$, 128.2, 127.5, 121.0, 61.5, 20.9, 14.5; IR (neat): ν_{max} 3361, 3256, 3218, 2929, 2853, 1689, 1605, 1363, 1186, 1047, 839, 755, 695 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + NH_4]^+$ calcd for $C_{22}H_{25}N_2O_5S$, 429.1479; found, 429.1482.

*Ethyl 5″-Bromo-2″-(sulfamoyloxy)-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**4ca**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **4ca** as a white solid. Yield, 67.4 mg (71%); mp 162−164 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.12 (d, *J* = 8.0 Hz, 1H), 8.06 (s, 1H), 7.54 (d, *J* = 8.0 Hz, 1H), 7.45−7.43 (m, 2H), 7.28−7.23 (m, 4H), 7.19−7.17 (m, 2H), 4.48 (s, 2H), 4.40 (q, *J* = 7.0, 2H), 1.42 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.5, 146.6, 146.0, 140.0, 135.5, 135.1, 134.3, 132.3, 130.8, 129.9, 129.6, 129.4, 128.4, 127.8, 122.9, 119.8, 61.6, 14.5; IR (neat): ν_{max} 3390, 3244, 3072, 2910, 2862, 1698, 1468, 1372, 1170, 1049, 857, 758, 699 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + NH_4]^+$ calcd for $C_{21}H_{22}{}^{79}BrN_2O_5S$ and $C_{21}H_{22}{}^{81}BrN_2O_5S$, 493.0427 and 495.0407; found, 493.0426 and 495.0406.

*Ethyl 2′-(Sulfamoyloxy)-6-(thiophen-2-yl)-[1,1′-biphenyl]-3-carboxylate (**4af**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2f** (63.8 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **4af** as a white solid. Yield, 52.4 mg (65%); mp 154−156 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.07−8.03 (m, 2H), 7.68 (d, *J* = 8.5 Hz, 1H), 7.48 (d, *J* = 8.5 Hz, 1H), 7.44−7.40 (m, 1H), 7.35−7.30 (m, 2H), 7.25−7.24 (m, 1H), 6.90−6.89 (m, 1H), 6.81−6.80 (m, 1H), 4.56 (s, 2H), 4.38 (q, *J* = 7.0 Hz, 2H), 1.39 (t, *J* = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.3, 148.1, 141.9, 138.4, 135.3, 133.4, 132.8, 132.4, 130.1, 129.9, 129.6, 129.4, 127.6$_4$, 127.5$_8$, 127.0$_5$, 127.0$_2$, 121.9, 61.5, 14.4; IR (neat): ν_{max} 3374, 3225, 3113, 2999, 2920, 2856, 1699, 1601, 1566, 1244, 1190, 1025, 768, 620 cm^{-1}; HRMS (ESI-TOF): *m/z* $[M + Na]^+$ calcd for $C_{19}H_{17}NNaO_5S_2$, 426.0440; found, 426.0439.

(iv) General Procedure for the Synthesis of Compounds 5aa−5da. A Schlenk tube was charged with cyclic sulfamidate imine **1** (0.20 mmol), triphenylphosphine (0.04 mmol), and toluene (1.0 mL). Subsequently, δ-acetoxy allenoate **2** (0.24 mmol) in toluene (1.0 mL) was added gradually over a period of 30 min at 80 °C (oil bath), the reaction mixture was stirred at the same temperature for the stipulated time, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by adding water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then, the combined organic layers were washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (10:90) as the eluent.

*Ethyl 2″-Hydroxy-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5aa**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg,

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J. Org. Chem. 2020, 85, 4130−4144

0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5aa** as a white solid. Yield, 51.5 mg (81%); mp 182−184 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.14 (dd, J = 8.0, 2.0 Hz, 1H), 8.10 (d, J = 2.0 Hz, 1H), 7.58 (d, J = 8.0 Hz, 1H), 7.25−7.16 (m, 6H), 7.06 (dd, J = 7.5, 1.5 Hz, 1H), 6.87 (td, J = 7.5, 1.0 Hz, 1H), 6.79 (dd, J = 8.0, 1.0 Hz, 1H), 4.86 (s, 1H), 4.40 (q, J = 7.0 Hz, 2H), 1.40 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.3, 152.5, 146.2, 139.9, 135.8, 132.7, 131.4, 130.8, 130.1, 129.7, 129.4, 129.1, 128.2, 127.7, 127.4, 120.8, 115.9, 61.3, 14.5; IR (neat): ν_{max} 3421, 3075, 2986, 2931, 2861, 1686, 1600, 1446, 1197, 1028, 749, 695 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{21}H_{22}NO_3$, 336.1594; found, 336.1590.

*Ethyl 2″-Hydroxy-4-methoxy-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5ab**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5ab** as a white solid. Yield, 52.2 mg (75%); mp 147−149 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.11 (dd, J = 8.0, 2.0 Hz, 1H), 8.07 (d, J = 1.5 Hz, 1H), 7.54 (d, J = 8.0 Hz, 1H), 7.19 (td, J = 8.0, 1.5 Hz, 1H), 7.13−7.11 (m, 2H), 7.09 (dd, J = 7.5, 1.5 Hz, 1H), 6.90 (td, J = 7.5, 1.0 Hz, 1H), 6.80 (dd, J = 8.0, 1.0 Hz, 1H), 6.77−6.75 (m, 2H), 4.81−4.78 (m, 1H), 4.39 (q, J = 7.0 Hz, 2H), 3.76 (s, 3H), 1.39 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.4, 159.3, 152.4, 145.6, 135.5, 132.9, 132.0, 131.3, 130.6, 130.3, 129.7, 129.6, 129.4, 127.5, 120.9, 116.0, 113.8, 61.3, 55.3, 14.5; IR (neat): ν_{max} 3390, 3075, 2992, 2934, 2839, 1706, 1607, 1445, 1175, 1033, 772, 688 cm^{-1}; HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{22}H_{21}O_4$, 349.1434; found, 349.1432.

*Ethyl 4-Chloro-2″-hydroxy-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5ac**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5ac** as a white solid. Yield, 58.4 mg (83%); mp 114−116 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.13 (dd, J = 8.0, 2.0 Hz, 1H), 8.09 (d, J = 1.5 Hz, 1H), 7.53 (d, J = 8.0 Hz, 1H), 7.21−7.18 (m, 3H), 7.12−7.09 (m, 2H), 7.04 (dd, J = 8.0, 2.0 Hz, 1H), 6.88 (td, J = 7.5, 1.0 Hz, 1H), 6.79 (dd, J = 8.5, 1.0 Hz, 1H), 4.95 (s, 1H), 4.39 (q, J = 7.0 Hz, 2H), 1.39 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.3, 152.5, 145.0, 138.5, 135.9, 133.8, 132.7, 131.3, 130.6, 130.4, 130.3, 129.7, 129.6, 128.4, 127.0, 120.9, 116.0, 61.4, 14.5; IR (neat): ν_{max} 3412, 3072, 2990, 2924, 2852, 1691, 1604, 1453, 1164, 1046, 769, 677 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{21}H_{21}{}^{35}ClNO_3$ and $C_{21}H_{21}{}^{37}ClNO_3$, 370.1204 and 372.1175; found, 370.1204 and 372.1179.

*Ethyl 2″-Hydroxy-4-nitro-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5ad**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2d** (73.2 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5ad** as a white solid. Yield, 62.0 mg (85%); mp 194−196 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.14 (dd, J = 8.0, 1.5 Hz, 1H), 8.12_4−8.12_1 (m, 1H), 8.06−8.04 (m, 2H), 7.55−7.53 (m, 1H), 7.35−7.32 (m, 2H), 7.20−7.17 (m, 1H), 7.04 (dd, J = 7.5, 1.5 Hz, 1H), 6.88 (td, J = 7.5, 1.0 Hz, 1H), 6.76 (dd, J = 8.0, 1.0 Hz, 1H), 4.98 (s, 1H), 4.40 (q, J = 7.0 Hz, 2H), 1.40 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.2, 152.4, 147.2, 147.0, 144.0, 136.4, 132.7, 131.3, 131.0, 130.4, 130.0, 129.9, 129.6, 126.5, 123.3, 121.1, 116.0, 61.6, 14.4; IR (neat): ν_{max} 3442, 3057, 2987, 2849, 1697, 1594, 1515, 1489, 1189, 1046, 776, 699 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{21}H_{21}N_2O_5$, 381.1445; found, 381.1442.

*tert-Butyl 3-(5-(Ethoxycarbonyl)-2′-hydroxy-[1,1′-biphenyl]-2-yl)-1H-indole-1-carboxylate (**5ae**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2e** (95.8 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5ae** as a gummy liquid. Yield, 59.4 mg (65%); ^1H NMR (500 MHz, $CDCl_3$): δ 8.18 (dd, J = 8.0, 2.0 Hz, 1H), 8.15−8.14 (m, 2H), 7.81 (d, J = 8.0 Hz, 1H), 7.56 (d, J = 8.0 Hz, 1H), 7.32−7.29 (m, 1H), 7.22−7.14 (m, 4H), 6.90 (td, J = 7.5, 1.0 Hz, 1H), 6.81 (dd, J = 8.0, 1.0 Hz, 1H), 4.78 (s, 1H), 4.42 (q, J = 7.0 Hz, 2H), 1.58 (s, 9H), 1.41 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.3, 152.5, 149.4, 138.0, 136.3, 135.5, 132.9, 131.1, 130.7, 129.9, 129.6, 129.5, 129.1, 127.5, 125.6, 124.7, 123.1, 120.9, 119.7, 118.9, 115.9, 115.4, 83.9, 61.3, 28.2, 14.5; IR (neat): ν_{max} 3418, 2980, 2926, 2856, 1716, 1604, 1451, 1369, 1237, 1155, 1070, 749 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{28}H_{31}N_2O_5$, 475.2227; found, 475.2233.

*Ethyl 2′-Hydroxy-6-(thiophen-2-yl)-[1,1′-biphenyl]-3-carboxylate (**5af**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1a** (39.4 mg, 0.20 mmol) with δ-acetoxy allenoate **2f** (63.8 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5af** as a pale yellow solid. Yield, 44.0 mg (68%); mp 141−143 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.10 (dd, J = 8.0, 2.0 Hz, 1H), 8.03 (d, J = 1.5 Hz, 1H), 7.75 (d, J = 8.5 Hz, 1H), 7.32−7.29 (m, 1H), 7.26−7.24 (m, 1H), 7.17 (dd, J = 7.5, 1.5 Hz, 1H), 6.99 (td, J = 7.5, 1.0 Hz, 1H), 6.92−6.88 (m, 3H), 4.72 (d, J = 4.0 Hz, 1H), 4.39 (q, J = 7.0 Hz, 2H), 1.39 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (500 MHz, $CDCl_3$): δ 166.1, 153.0, 141.1, 138.6, 134.7, 133.2, 131.3, 130.0, 129.8_9, 129.8_8, 129.8_0, 127.5_1, 127.4_7, 127.4_1, 127.1, 121.1, 116.0, 61.4, 14.5; IR (neat): ν_{max} 3403, 3117, 3078, 2984, 2922, 2848, 1684, 1445, 1304, 1253, 1198, 1027, 771, 692 cm^{-1}; HRMS (ESI-TOF): m/z $[M + H]^+$ calcd for $C_{19}H_{17}O_3S$, 325.0893; found, 325.0896.

*Ethyl 2″-Hydroxy-5″-methyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5ba**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5ba** as a white solid. Yield, 49.1 mg (74%); mp 143−145 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.13 (dd, J = 8.0, 2.0 Hz, 1H), 8.08 (d, J = 1.5 Hz, 1H), 7.56 (d, J = 8.5 Hz, 1H), 7.26−7.20 (m, 5H), 6.97 (dd, J = 8.5, 2.0 Hz, 1H), 6.91 (d, J = 2.0 Hz, 1H), 6.65 (d, J = 7.5 Hz, 1H), 4.53−4.52 (m, 1H), 4.40 (q, J = 7.0 Hz, 2H), 2.23 (s, 3H), 1.41 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.4, 150.2, 146.0, 140.0, 136.1, 132.8, 131.7, 130.7, 130.1, 130.0, 129.9, 129.5, 129.1, 128.3, 127.7, 127.2, 115.8, 61.3, 20.6, 14.5; IR (neat): ν_{max} 3365, 3060, 3024, 2972, 2921, 2859, 1692, 1601, 1471, 1264, 1126, 1044, 757, 701 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{22}H_{24}NO_3$, 350.1751; found, 350.1750.

*Ethyl 2″-Hydroxy-4-methoxy-5″-methyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5bb**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5bb** as a white solid. Yield, 50.7 mg (70%); mp 139−141 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.10 (d, J = 8.0 Hz, 1H), 8.05 (s, 1H), 7.53 (d, J = 8.0 Hz, 1H), 7.14 (d, J = 8.5 Hz, 2H), 6.99 (d, J = 8.5 Hz, 1H), 6.95 (s, 1H), 6.77 (d, J = 8.5 Hz, 2H), 6.67 (d, J = 8.0 Hz, 1H), 4.48 (s, 1H), 4.39 (q, J = 7.0 Hz, 2H), 3.77 (s, 3H), 2.26 (s, 3H), 1.40 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.4, 159.4, 150.1, 145.4, 135.8, 133.0, 132.1, 131.6, 130.5, 130.3, 130.1, 129.9, 129.7, 129.6, 127.4, 115.9, 113.9, 61.2, 55.3, 20.6, 14.5; IR (neat): ν_{max} 3430, 2925, 2854, 1707, 1602, 1557, 1459, 1241, 1171, 1032, 768, 687 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{23}H_{26}NO_4$, 380.1856; found, 380.1857.

*Ethyl 4-Chloro-2″-hydroxy-5″-methyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5bc**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5bc** as a white solid. Yield, 55.0 mg (75%); mp 155−157 °C; ^1H NMR (500 MHz, $CDCl_3$): δ 8.10 (dd, J = 8.0, 1.5 Hz, 1H), 8.07 (d, J = 1.5 Hz, 1H), 7.50 (d, J = 8.0 Hz, 1H), 7.21−7.18 (m, 2H), 7.14−7.12 (m, 2H), 6.98 (dd, J = 8.5, 2.0 Hz, 1H), 6.88 (d, J = 2.0 Hz, 1H), 6.66 (d, J = 8.0 Hz, 1H), 4.74−4.72 (m, 1H), 4.39 (q, J = 7.0 Hz, 2H), 2.24 (s, 3H), 1.40 (t, J = 7.0 Hz, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 166.3, 150.1, 144.9, 138.6, 136.1, 133.8, 132.8, 131.6, 130.5, 130.4, 130.3, 130.1_4, 130.1_0, 129.6, 128.4, 126.8, 115.8, 61.4, 20.6, 14.5; IR (neat): ν_{max} 3382, 3063, 2988, 2867, 1691, 1603, 1471, 1273, 1090, 1048, 774, 679 cm^{-1}; HRMS (ESI-TOF): m/z $[M + NH_4]^+$ calcd for $C_{22}H_{23}{}^{35}ClNO_3$ and $C_{22}H_{23}{}^{37}ClNO_3$, 384.1361 and 386.1331; found, 384.1362 and 386.1339.

*Ethyl 2″-Hydroxy-5″-methyl-4-nitro-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5bd**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1b** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2d** (73.2 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol)

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J. Org. Chem. 2020, 85, 4130−4144

afforded **5bd** as a white solid. Yield, 58.0 mg (77%); mp 217−219 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.13 (dd, J = 8.0, 1.5 Hz, 1H), 8.10 (m, 1H), 8.06 (d, J = 8.5 Hz, 2H), 7.53 (d, J = 8.0 Hz, 1H), 7.35 (d, J = 8.5 Hz, 2H), 6.98 (dd, J = 8.0, 2.0 Hz, 1H), 6.89 (s, 1H), 6.62 (d, J = 8.0 Hz, 1H), 4.69 (s, 1H), 4.40 (q, J = 7.0 Hz, 2H), 2.23 (s, 3H), 1.41 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.1, 150.1, 147.3, 147.1, 143.9, 136.6, 132.8, 131.6, 131.0, 130.3$_9$, 130.3$_6$, 130.2$_8$, 130.0, 129.5, 126.4, 123.3, 115.8, 61.5, 20.5, 14.5; IR (neat): ν_{max} 3423, 3120, 3045, 2988, 2851, 1698, 1596, 1492, 1295, 1189, 1046, 751, 696 cm^{-1}; HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{22}H_{20}NO_5$, 378.1336; found, 378.1337.

*Ethyl 5″-Bromo-2″-hydroxy-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5ca**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5ca** as a white solid. Yield, 58.0 mg (73%); mp 182−184 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.14 (dd, J = 8.0, 2.0 Hz, 1H), 8.04 (d, J = 1.5 Hz, 1H), 7.56 (d, J = 8.0 Hz, 1H), 7.28−7.25 (m, 5H), 7.20−7.18 (m, 2H), 6.64 (d, J = 9.0 Hz, 1H), 4.77 (s, 1H), 4.40 (q, J = 7.0 Hz, 2H), 1.41 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.1, 151.7, 146.1, 139.4, 134.4, 133.7, 132.6, 132.2, 130.9, 130.3, 130.1, 129.5, 129.0, 128.5, 128.1, 117.8, 112.7, 61.4, 14.5; IR (neat): ν_{max} 3335, 3063, 3024, 2981, 2931, 2869, 1687, 1600, 1561, 1462, 1201, 1041, 756, 702 cm^{-1}; HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{21}H_{18}{}^{79}BrO_3$ and $C_{21}H_{18}{}^{81}BrO_3$, 397.0434 and 399.0413; found, 397.0436 and 399.0418.

*Ethyl 5″-Bromo-2″-hydroxy-4-methoxy-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5cb**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5cb** as a gummy liquid. Yield, 59.0 mg (69%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.12 (dd, J = 8.0, 1.5 Hz, 1H), 8.01 (d, J = 1.5 Hz, 1H), 7.53 (d, J = 8.0 Hz, 1H), 7.29−7.27 (m, 2H), 7.14−7.12 (m, 2H), 6.80−6.78 (m, 2H), 6.67−6.65 (m, 1H), 4.81 (s, 1H), 4.39 (q, J = 7.0 Hz, 2H), 3.78 (s, 3H), 1.40 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.2, 159.6, 151.6, 145.5, 134.2, 133.6, 132.8, 132.2, 131.6, 130.7, 130.2, 130.1, 129.8$_3$, 129.7$_7$, 117.9, 114.0, 112.8, 61.4, 55.3, 14.5; IR (neat): ν_{max} 3457, 3063, 2964, 1725, 1601, 1515, 1464, 1264, 1172, 731, 703 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{22}H_{23}{}^{79}BrNO_4$ and $C_{22}H_{23}{}^{81}BrNO_4$, 444.0805 and 446.0785; found, 444.0812 and 446.0795.

*Ethyl 5″-Bromo-4-chloro-2″-hydroxy-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5cc**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5cc** as a white solid. Yield, 63.0 mg (73%); mp 179−181 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.11 (dd, J = 8.0, 2.0 Hz, 1H), 8.02 (d, J = 2.0 Hz, 1H), 7.50 (d, J = 8.0 Hz, 1H), 7.28 (dd, J = 8.5, 2.0 Hz, 1H), 7.23−7.21 (m, 3H), 7.12−7.10 (m, 2H), 6.65 (d, J = 9.0 Hz, 1H), 5.05 (s, 1H), 4.38 (q, J = 7.0 Hz, 2H), 1.40 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.1, 151.7, 144.9, 138.1, 134.6, 134.1, 133.6, 132.6, 132.4, 130.6, 130.4, 130.3, 130.1, 129.2, 128.6, 117.8, 112.8, 61.5, 14.5; IR (neat): ν_{max} 3310, 3072, 2977, 2933, 2864, 1683, 1601, 1473, 1275, 1191, 1041, 772, 684 cm^{-1}; HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{21}H_{17}{}^{35}Cl^{79}BrO_3$, $C_{21}H_{17}{}^{35}Cl^{81}BrO_3$/$C_{21}H_{17}{}^{37}Cl^{79}BrO_3$, and $C_{21}H_{17}{}^{37}Cl^{81}BrO_3$, 431.0044, 433.0024, and 434.9990; found 431.0048, 433.0031, and 434.9999.

*Ethyl 5″-Bromo-2″-hydroxy-4-nitro-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5cd**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1c** (55.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2d** (73.2 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5cd** as a pale yellow solid. Yield, 67.0 mg (76%); mp 210−212 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.14 (dd, J = 8.0, 1.5 Hz, 1H), 8.10−8.06 (m, 3H), 7.53 (d, J = 8.0 Hz, 1H), 7.35 (d, J = 9.0 Hz, 2H), 7.29 (dd, J = 9.0, 2.5 Hz, 1H), 7.25 (d, J = 2.5 Hz, 1H), 6.63 (d, J = 9.0 Hz, 1H), 5.05 (s, 1H), 4.40 (q, J = 7.0 Hz, 2H), 1.41 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.0, 151.6, 147.3, 146.7, 143.9, 135.0, 133.7, 132.7, 132.6, 131.2, 130.5, 130.1, 129.9, 128.7, 123.5, 117.8, 113.0, 61.7, 14.5; IR (neat): ν_{max} 3452, 3071, 2985, 2936, 2840, 1698, 1595, 1257, 1186, 1041, 746, 695 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{21}H_{20}{}^{79}BrN_2O_5$ and $C_{21}H_{20}{}^{81}BrN_2O_5$, 459.0550 and 461.0530; found, 459.0549 and 461.0540.

*Ethyl 2″-Hydroxy-3′-methyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**5da**).* Following the general procedure, the reaction of cyclic sulfamidate imine **1d** (42.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol) and Ph_3P (10.5 mg, 0.04 mmol) afforded **5da** as a white solid. Yield, 48.0 mg (72%); mp 113−115 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.92 (d, J = 8.0 Hz, 1H), 7.37 (d, J = 8.0 Hz, 1H), 7.17−7.15 (m, 3H), 7.13−7.11 (m, 1H), 7.09−7.07 (m, 2H), 6.83−6.81 (m, 2H), 6.79−6.76 (m, 1H), 4.68 (s, 1H), 4.41 (q, J = 7.0 Hz, 2H), 2.23 (s, 3H), 1.42 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 168.3, 152.9, 146.5, 140.8, 139.3, 136.0, 131.3, 131.2, 130.1, 129.3, 129.0, 127.9, 127.8, 127.2, 126.2, 120.6, 115.3, 61.2, 18.2, 14.5; IR (neat): ν_{max} 3420, 3063, 2980, 2929, 2853, 1714, 1585, 1490, 1170, 1070, 755, 700 cm^{-1}; HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{22}H_{21}O_3$, 333.1485; found, 333.1483.

(v) Scale-Up Experiments for 3ab and 5ab. *Synthesis of 2-Pyridinyl Acetate **3ab**.* A 25 mL RB flask was charged with cyclic sulfamidate imine **1a** (197.0 mg, 1.0 mmol), δ-acetoxy allenoate **2b** (348.0 mg, 1.2 mmol), Na_2CO_3 (159.0 mg, 1.5 mmol), and 8.0 mL of toluene. Subsequently, DBU (75 μL, 0.5 mmol) in 2.0 mL of toluene was added, and the mixture stirred at 50 °C (oil bath) for 12 h. After completion of the reaction, it was quenched by adding water (30 mL). The aqueous layer was extracted with ethyl acetate (3 × 15 mL). Then, the combined organic layer was washed with brine (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by silica gel column chromatography using ethyl acetate/hexane (10:90) as the eluent to obtain **3ab** (261.0 mg) in 72% yield.

*Synthesis of Teraryl Motif **5ab**.* A 25 mL RB flask was charged with cyclic sulfamidate imine **1a** (197.0 mg, 1.0 mmol), Ph_3P (52.5 mg, 0.20 mmol), and toluene (5.0 mL). Subsequently, δ-acetoxy allenoate **2b** (348.0 mg, 1.20 mmol) in toluene (5.0 mL) was added gradually over a period of 45 min at 80 °C (oil bath), and the mixture was stirred at the same temperature for 12 h. Then, it was quenched by adding water (30 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then, the combined organic layer was washed with brine (20 mL), dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was purified by silica gel column chromatography using ethyl acetate/hexane (10:90) as the eluent to obtain **5ab** (254.0 mg) in 73% yield.

(vi) General Procedure for the Synthesis of Compounds 7aa−7bc. A Schlenk tube was charged with cyclic sulfonyl imine **6** (0.20 mmol), δ-acetoxy allenoate **2** (0.24 mmol), Na_2CO_3 (0.30 mmol), and toluene (1.5 mL). Subsequently, DBU (0.20 mmol) in toluene (0.5 mL) was added, the mixture was stirred at 100 °C (oil bath) for the stipulated time, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by adding water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then, the combined organic layer was washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (5:95) as the eluent.

*Ethyl 2-(4,6-Diphenylpyridin-2-yl)acetate (**7aa**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7aa** as a gummy liquid. Yield, 45.0 mg (71%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.06−8.04 (m, 2H), 7.82 (d, J = 1.5 Hz, 1H), 7.70−7.69 (m, 2H), 7.52−7.40 (m, 7H), 4.23 (q, J = 7.0 Hz, 2H), 3.99 (s, 2H), 1.30 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.0, 157.9, 155.0, 150.1, 139.6, 138.7, 129.2, 129.1$_5$, 129.1$_2$, 128.8, 127.3$_2$, 127.2$_9$, 120.4, 117.3, 61.1, 44.4, 14.4; IR (neat): ν_{max} 3069, 2920, 2856, 1733, 1596, 1550, 1436, 1251, 1154, 1030, 762, 694 cm^{-1}; HRMS (ESI-TOF): m/z [M + H]$^+$ calcd for $C_{21}H_{20}NO_2$, 318.1489; found, 318.1494.

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J. Org. Chem. 2020, 85, 4130−4144

*Ethyl 2-(4-(4-Methoxyphenyl)-6-phenylpyridin-2-yl)acetate (**7ab**).* Following the general procedure, the reaction of cyclic sulfonyl imine imine **6a** (36.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7ab** as a gummy liquid. Yield, 48.0 mg (69%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.04 (d, *J* = 7.0 Hz, 2H), 7.79 (d, *J* = 1.0 Hz, 1H), 7.65 (d, *J* = 9.0 Hz, 2H), 7.49−7.46 (m, 2H), 7.43−7.40 (m, 2H), 7.02 (d, *J* = 9.0 Hz, 2H), 4.23 (q, *J* = 7.0 Hz, 2H), 3.97 (s, 2H), 3.87 (s, 3H), 1.30 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.0, 160.7, 157.8, 155.0, 149.5, 139.7, 131.0, 129.0, 128.8, 128.4, 127.3, 119.8, 116.8, 114.7, 61.1, 55.5, 44.4, 14.4; IR (neat): ν_{max} 3040, 2920, 2856, 1731, 1599, 1514, 1430, 1249, 1177, 1028, 775, 693 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{22}NO_3$, 348.1594; found, 348.1596.

*Ethyl 2-(4-(4-Chlorophenyl)-6-phenylpyridin-2-yl)acetate (**7ac**).* Following the general procedure, the reaction of cyclic sulfonyl imine imine **6a** (36.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7ac** as a gummy liquid. Yield, 51.2 mg (73%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.04−8.02 (m, 2H), 7.77 (d, *J* = 1.5 Hz, 1H), 7.64−7.61 (m, 2H), 7.50−7.46 (m, 4H), 7.44−7.41 (m, 2H), 4.23 (q, *J* = 7.0 Hz, 2H), 3.99 (s, 2H), 1.30 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.0, 158.1, 155.2, 148.8, 139.3, 137.2, 135.4, 129.4, 129.2, 128.9, 128.6, 127.3, 120.2, 117.1, 61.2, 44.3, 14.4; IR (neat): ν_{max} 3031, 2922, 2862, 1733, 1602, 1546, 1255, 1153, 1029, 824, 774, 693 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{21}H_{19}{}^{35}ClNO_2$ and $C_{21}H_{19}{}^{37}ClNO_2$, 352.1099 and 354.1069; found, 352.1100 and 354.1069.

*Ethyl 2-(5-Methyl-4,6-diphenylpyridin-2-yl)acetate (**7ba**).* Following the general procedure, the reaction of cyclic sulfonyl imine imine **6b** (39.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7ba** as a gummy liquid. Yield, 45.7 mg (69%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.55−7.53 (m, 2H), 7.47−7.43 (m, 4H), 7.42−7.36 (m, 4H), 7.18 (s, 1H), 4.19 (q, *J* = 7.0 Hz, 2H), 3.90 (s, 2H), 2.17 (s, 3H), 1.27 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.2, 159.5, 151.5, 151.1, 141.1, 139.9, 129.3, 128.9, 128.5, 128.3, 128.0, 127.0, 123.4, 61.0, 43.7, 17.9, 14.3; IR (neat): ν_{max} 3069, 2923, 2859, 1733, 1585, 1544, 1264, 1151, 1029, 734, 700 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{22}NO_2$, 332.1645; found, 332.1648.

*Ethyl 2-(4-(4-Methoxyphenyl)-5-methyl-6-phenylpyridin-2-yl)-acetate (**7bb**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6b** (39.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7bb** as a gummy liquid. Yield, 46.2 mg (64%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.55−7.53 (m, 2H), 7.46−7.43 (m, 2H), 7.39−7.36 (m, 1H), 7.33−7.31 (m, 2H), 7.17 (s, 1H), 7.00−6.98 (m, 2H), 4.20 (q, *J* = 7.0 Hz, 2H), 3.90 (s, 2H), 3.86 (s, 3H), 2.19 (s, 3H), 1.27 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.1, 159.5$_7$, 159.4$_9$, 151.1$_2$, 151.0$_9$, 141.3, 132.2, 130.2, 129.3, 128.3, 128.0, 127.0, 123.5, 113.9, 61.0, 55.4, 43.7, 18.0, 14.3; IR (neat): ν_{max} 2923, 2853, 1735, 1598, 1511, 1249, 1175, 1032, 845, 714 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{23}H_{24}NO_3$, 362.1751; found, 362.1753.

*Ethyl 2-(4-(4-Chlorophenyl)-5-methyl-6-phenylpyridin-2-yl)-acetate (**7bc**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6b** (39.0 mg, 0.20 mmol) with δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **3ad** as a gummy liquid. Yield, 51.1 mg (70%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.53−7.52 (m, 2H), 7.46−7.42 (m, 4H), 7.40−7.37 (m, 1H), 7.32−7.31 (m, 2H), 7.15 (s, 1H), 4.19 (q, *J* = 7.0 Hz, 2H), 3.90 (s, 2H), 2.16 (s, 3H), 1.27 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.0, 159.7, 151.3, 150.2, 141.0, 138.4, 134.2, 130.3, 129.3, 128.8, 128.3, 128.1, 126.9, 123.2, 61.0, 43.6, 17.8, 14.3; IR (neat): ν_{max} 3058, 2929, 2853, 1732, 1542, 1490, 1383, 1150, 1029, 831, 786, 700 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{21}{}^{35}ClNO_2$ and $C_{22}H_{21}{}^{37}ClNO_2$, 366.1255 and 368.1226; found, 366.1259 and 368.1232.

*tert-Butyl 3-(2-(2-Ethoxy-2-oxoethyl)-6-phenylpyridin-4-yl)-1H-indole-1-carboxylate (**7ae**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2e** (95.8 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7ae** as a gummy liquid. Yield, 52.0 mg (57%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.26 (d, *J* = 8.0 Hz, 1H), 8.07−8.05 (m, 2H), 7.91−7.89 (m, 3H), 7.53 (d, *J* = 1.0 Hz, 1H), 7.51−7.48 (m, 2H), 7.45−7.40 (m, 2H), 7.37−7.33 (m, 1H), 4.25 (q, *J* = 7.0 Hz, 2H), 4.01 (s, 2H), 1.72 (s, 9H), 1.32 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 171.0, 157.9, 155.1, 149.6, 143.2, 139.6, 136.2, 129.2, 128.9, 128.3, 127.3, 125.2, 124.5, 123.5, 120.9$_0$, 119.9$_6$, 119.9$_7$, 117.8, 115.8, 84.6, 61.1, 44.4, 28.4, 14.4; IR (neat): ν_{max} 3066, 2980, 2926, 2853, 1731, 1605, 1451, 1235, 1148, 1028, 765, 694 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{28}H_{29}N_2O_4$, 457.2122; found, 457.2126.

*Ethyl 2-(6-Phenyl-4-(thiophen-2-yl)pyridin-2-yl)acetate (**7af**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol) with δ-acetoxy allenoate **2f** (63.8 mg, 0.24 mmol), DBU (30.4 mg, 0.10 mmol), and Na_2CO_3 (31.8 mg, 0.30 mmol) afforded **7af** as a gummy liquid. Yield, 40.0 mg (62%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.04−8.02 (m, 2H), 7.79 (d, *J* = 1.5 Hz, 1H), 7.56 (dd, *J* = 3.5, 1.0 Hz, 1H), 7.50−7.41 (m, 5H), 7.15 (dd, *J* = 5.0, 3.5 Hz, 1H), 4.23 (q, *J* = 7.0 Hz, 2H), 3.95 (s, 2H), 1.30 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 170.8, 158.1, 155.3, 143.0, 141.7, 139.4, 129.2, 128.8, 128.5, 127.3, 127.1, 125.5, 118.6, 115.6, 61.1, 44.3, 14.4; IR (neat): ν_{max} 3078, 2979, 2925, 2850, 1731, 1597, 1551, 1421, 1178, 1028, 733, 694 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{19}H_{18}NO_2S$, 324.1053; found, 324.1055.

(vii) Synthesis of Ethyl 2-(9-(4-Methoxyphenyl)-10-methyl-5,5-dioxido-9*H*-benzo[4,5]isothiazolo[2,3-*a*]pyridin-7-yl)-acetate 8bb. A Schlenk tube was charged with 3-ethylbenzo[*d*]-isothiazole 1,1-dioxide **6b** (39.0 mg, 0.20 mmol), ethyl 5-acetoxy-5-(4-methoxyphenyl)penta-2,3-dienoate **2b** (69.6 mg, 0.24 mmol), and toluene (1.5 mL). Subsequently, DBU (46.0 mg, 0.30 mmol) in toluene (0.5 mL) was added, the mixture was stirred at 60 °C (oil bath) for 12 h, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by adding water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then, the combined organic layer was washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product **8bb** was then purified by silica gel column chromatography using ethyl acetate/hexane (10:90) as the eluent. White solid; yield, 63.0 mg (74%); mp 136−139 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.84 (d, *J* = 8.0 Hz, 1H), 7.80 (d, *J* = 8.0 Hz, 1H), 7.64−7.61 (m, 1H), 7.54−7.51 (m, 1H), 7.25 (d, *J* = 8.5 Hz, 2H), 6.86 (d, *J* = 9.0 Hz, 2H), 4.96 (d, *J* = 4.0 Hz, 1H), 4.22−4.18 (m, 2H), 4.06 (d, *J* = 4.0 Hz, 1H), 3.78 (s, 3H), 3.76−3.62 (m, 2H), 2.00 (s, 3H), 1.28 (t, *J* = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): 169.8, 159.0, 136.0, 133.1, 132.7, 130.0, 129.7, 129.0, 125.2, 124.7, 124.1, 121.4, 117.5, 114.3, 111.8, 61.3, 55.4, 47.7, 37.4, 18.2, 14.3; IR (neat): ν_{max} 2967, 2920, 2840, 1734, 1601, 1509, 1293, 1022, 759, 648 cm^{-1}; HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{23}H_{24}NO_5S$, 426.1370; found, 426.1352.

(viii) General Procedure for the Synthesis of Compounds 9aa−9ba. A Schlenk tube was charged with cyclic sulfonyl imine **6** (0.20 mmol), triphenylphosphine (0.04 mmol), and toluene (1.0 mL). Subsequently, δ-acetoxy allenoate **2** (0.24 mmol) in toluene (1.0 mL) was added gradually over 30 min at 80 °C (oil bath), the mixture was stirred at the same temperature for the stipulated time, and the progress of the reaction was monitored using TLC. After completion of the reaction, the mixture was quenched by adding water (10 mL). The aqueous layer was extracted with ethyl acetate (3 × 5 mL). Then, the combined organic layers were washed with brine, dried over anhydrous Na_2SO_4, and concentrated under reduced pressure. The crude product was then purified by silica gel column chromatography using ethyl acetate/hexane (30:70) as the eluent.

*Ethyl 2″-Sulfamoyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**9aa**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol), δ-acetoxy allenoate **2a** (62.4 mg, 0.24

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J. Org. Chem. 2020, 85, 4130−4144

mmol) and triphenylphosphine (10.5 mg, 0.04 mmol) afforded **9aa** as a white solid. Yield, 58.0 mg (76%); mp 183−185 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.15 (dd, J = 8.0, 2.0 Hz, 1H), 8.11 (d, J = 2.0 Hz, 1H), 8.06−8.04 (m, 1H), 7.57 (d, J = 8.0 Hz, 1H), 7.45−7.41 (m, 2H), 7.25−7.20 (m, 5H), 7.15−7.13 (m, 1H), 4.42−4.36 (m, 2H), 4.31 (s, 2H), 1.39 (t, J = 7.0, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): 166.3, 145.2, 140.7, 139.9, 139.2, 138.1, 133.4, 132.1, 131.2, 130.6, 129.7$_2$, 129.6$_9$, 129.0, 128.5, 128.3, 128.2, 127.6, 61.4, 14.5; IR (neat): ν_{max} 3421, 3270, 3078, 2990, 2906, 2360, 1683, 1605, 1364, 1265, 1157, 1037, 757, 688 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{21}H_{23}N_2O_4S$, 399.1373; found, 399.1376.

*Ethyl 4-Methoxy-2″-sulfamoyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**9ab**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol), δ-acetoxy allenoate **2b** (69.6 mg, 0.24 mmol), and triphenylphosphine (10.5 mg, 0.04 mmol) afforded **9ab** as a white solid. Yield, 58.4 mg (71%); mp 148−150 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.12 (dd, J = 8.0, 2.0 Hz, 1H), 8.08−8.05 (m, 2H), 7.54 (d, J = 8.0 Hz, 1H), 7.46−7.41 (m, 2H), 7.16−7.13 (m, 3H), 6.73 (d, J = 8.5 Hz, 2H), 4.41−4.35 (m, 4H), 3.74 (s, 3H), 1.39 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.4, 159.1, 144.8, 140.7, 139.5, 137.9, 133.4, 132.2, 131.3, 130.9, 130.5, 129.8, 128.6, 128.5, 128.2, 113.7, 61.3, 55.3, 14.5; IR (neat): ν_{max} 3317, 3218, 3107, 2973, 2908, 2834, 1703, 1609, 1526, 1242, 1166, 1039, 770 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{22}H_{25}N_2O_5S$, 429.1479; found, 429.1478.

*Ethyl 4-Chloro-2″-sulfamoyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**9ac**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol), δ-acetoxy allenoate **2c** (70.6 mg, 0.24 mmol), and triphenylphosphine (10.5 mg, 0.04 mmol) afforded **9ac** as a white solid. Yield, 65.6 mg (79%); mp 208−210 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.14−8.11 (m, 2H), 8.08−8.05 (m, 1H), 7.52 (d, J = 8.0 Hz, 1H), 7.44−7.38 (m, 2H), 7.19−7.15 (m, 4H), 7.05−7.02 (m, 1H), 4.63 (s, 2H), 4.41−4.35 (m, 2H), 1.39 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.2, 144.4, 141.0, 138.9, 138.4, 138.1, 133.8, 133.2, 132.2, 131.1, 130.9, 130.6, 129.8, 129.3, 128.5, 128.4, 128.3, 61.5, 14.5; IR (neat): ν_{max} 3431, 3282, 3072, 2993, 2910, 1687, 1607, 1483, 1345, 1262, 1158, 1016, 751, 701 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{21}H_{22}{}^{35}ClN_2O_4S$ and $C_{21}H_{22}{}^{37}ClN_2O_4S$, 433.0983 and 435.0954; found, 433.0987 and 435.0962.

*Ethyl 4-Nitro-2″-sulfamoyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**9ad**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol), δ-acetoxy allenoate **2d** (73.2 mg, 0.24 mmol), and triphenylphosphine (10.5 mg, 0.04 mmol) afforded **9ad** as a pale yellow solid. Yield, 70.0 mg (82%); mp 197−199 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.18−8.16 (m, 2H), 8.10−8.09 (m, 1H), 8.04 (d, J = 8.0 Hz, 2H), 7.56 (d, J = 8.0 Hz, 1H), 7.44−7.37 (m, 4H), 7.00 (d, J = 7.5 Hz, 1H), 4.78−4.73 (m, 2H), 4.42−4.38 (m, 2H), 1.40 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.1, 147.2, 146.7, 143.5, 141.1, 138.3$_1$, 138.3$_0$, 133.0, 132.2, 131.2, 130.6, 130.5, 130.1, 129.8, 128.6, 128.5, 123.3, 61.6, 14.4; IR (neat): ν_{max} 3336, 3255, 2980, 2850, 1695, 1596, 1516, 1259, 1159, 1032, 775, 697 cm^{-1}; HRMS (ESI-TOF): m/z [M + Na]$^+$ calcd for $C_{21}H_{18}N_2NaO_6S$, 449.0778; found, 449.0779.

*Ethyl 2′-Sulfamoyl-6-(thiophen-2-yl)-[1,1′-biphenyl]-3-carboxylate (**9af**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (36.2 mg, 0.20 mmol), δ-acetoxy allenoate **2f** (63.8 mg, 0.24 mmol), and triphenylphosphine (10.5 mg, 0.04 mmol) afforded **9af** as a gummy liquid. Yield, 54.2 mg (70%); ^{1}H NMR (500 MHz, $CDCl_3$): δ 8.12−8.09 (m, 2H), 8.04 (m, 1H), 7.74 (d, J = 8.5 Hz, 1H), 7.58−7.52 (m, 2H), 7.28 (dd, J = 7.0, 1.5 Hz, 1H), 7.21 (dd, J = 4.5, 1.5 Hz, 1H), 6.89−6.87 (m, 2H), 4.39−4.33 (m, 4H), 1.37 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 166.1, 141.6, 141.0, 138.9, 137.5, 136.9, 133.1, 132.6, 132.2, 129.9, 129.4, 129.0, 128.9, 128.6, 127.8, 127.5$_4$, 127.5$_0$, 61.4, 14.4; IR (neat): ν_{max} 3383, 3256, 2964, 2922, 2862, 1707, 1607, 1307, 1260, 1162, 1029, 768, 701 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{19}H_{21}N_2O_4S_2$, 405.0937; found, 405.0944.

*Ethyl 3′-Methyl-2″-sulfamoyl-[1,1′:2′,1″-terphenyl]-4′-carboxylate (**9ba**).* Following the general procedure, the reaction of cyclic sulfonyl imine **6a** (39.0 mg, 0.20 mmol), δ-acetoxy allenoate **2a** (62.4 mg, 0.24 mmol), and triphenylphosphine (10.5 mg, 0.04 mmol) afforded **9ba** as a white solid. Yield, 54.5 mg (69%); mp 154−156 °C; ^{1}H NMR (500 MHz, $CDCl_3$): δ 7.96−7.94 (m, 2H), 7.56 (td, J = 7.5, 1.0 Hz, 1H), 7.44 (td, J = 8.0, 1.0 Hz, 1H), 7.34−7.32 (m, 2H), 7.18−7.13 (m, 5H), 4.38 (q, J = 7.0 Hz, 2H), 4.10 (s, 2H), 2.25 (s, 3H), 1.41 (t, J = 7.0 Hz, 3H); ^{13}C{^{1}H} NMR (125 MHz, $CDCl_3$): δ 168.1, 144.1, 140.8, 140.4, 138.9, 138.3, 134.0, 132.0, 130.7, 130.3, 129.9, 128.7, 128.5, 128.0, 127.6, 127.4, 61.2, 19.5, 14.5; IR (neat): ν_{max} 3428, 3272, 3081, 2990, 2915, 2876, 1687, 1602, 1301, 1157, 1016, 773, 700 cm^{-1}; HRMS (ESI-TOF): m/z [M + NH_4]$^+$ calcd for $C_{22}H_{25}N_2O_4S$, 413.1530; found, 413.1533.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.9b03281.

ORTEPs of compounds **3aa**, **3db**, **4af**, **4ca**, **5aa**, **5bd**, **8bb**, and **9aa** (Figures S1−S8; CCDC: 1952241−1952248), Table S1, HRMS data for intermediates, and ^{1}H/^{13}C NMR spectra (PDF)

X-ray crystallographic data (CIF)

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Notes

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